

- **Third trimester**

The third trimester represents a period of rapid fetal growth. During the early stage of this period, most of the organ systems become fully functional. During the third trimester, the fetus grows to about 50 cm in length and weighs about 3-4 kg. Fetal activity decrease as it fills the space within the uterus.

Maternal immune tolerances of the embryo and fetus

During pregnancy the overall regulation of the mother's immune system changes. These changes allow the mother to keep the embryo in her uterus without rejecting as a foreign body even though half the embryo's genes are inherited from the father and many chemical markers on the surface of the embryo are foreign to the mother.

Process of parturition

Child birth begins with the labor. The labor is a series of strong, rhythmic uterine contractions that push the fetus and placenta out of the body. When labor begins, local regulators (prostaglandins) and hormones (mainly estradiol and oxytocin) induce and regulate further contractions of the uterus. This is a positive feedback mechanism as uterine contractions stimulate secretion of oxytocin which stimulates further contractions of the uterus.

The labor can be divided into three stages. The first stage is the thinning and opening up (dilation) of the cervix. The second stage is the delivery of the baby. In this stage, continuous and strong contractions force the fetus out of the uterus and expell through the vagina. Delivery of the placenta is the final stage of labor.

Lactation

The lactation which is unique to mammals begins as post natal care. Lactation is the secretion and ejection of mother's milk from the mammary glands. Lactation is subjected to nervous and hormonal regulation. The main hormone in promoting milk synthesis and secretion is prolactin. In response to suckling by the new born baby (which initiates nerve impulses from touch receptors in the nipples) and decrease in estradiol and progesterone levels in the mother's blood after birth, the hypothalamus send impulses to the anterior pituitary to secrete prolactin hormone which stimulates the mammary glands to produce milk. Suckling also stimulates the secretion of oxytocin hormone from the posterior pituitary gland and triggers the release (ejection) of milk from the mammary glands. This is a positive feedback mechanism as milk availability encourages continuous suckling, so touch sensation on the nipple and oxytocin release continue further ejecting milk from the mammary glands.

Composition of human milk and significance of breast feeding

During the first few days after birth, the mammary glands secrete a fluid called 'colostrum' until appearance of true milk. Human milk is a sterile solution that contains nutrients such as lactose, fatty acids, amino acids, minerals, vitamins, and water that are ideal for baby's digestion, brain development and growth. Human milk includes proteins such as casein, lactalbumin and immunoglobulins.

Colostrum and true milk provide nutrients for the baby and contain important antibodies that protect the infant. Several types of white blood cells are also present in the breast milk which help the baby to resist microbial infections. Compared to the true milk, colostrum contains less nutrients (less lactose and no fat) but they are adequate for the early nutritional needs.

Breast feeding supports optimal infant growth, enhances intellectual development and fosters mother-infant relations by establishing early and prolonged contact between them. Compared to cow's milk, the fat, iron and the proteins in the breast milk are more readily metabolized. Lower sodium content of breast milk is more suited to the baby's needs. The baby is less likely to have allergic reactions to mother's milk than the milk from another source.

Birth control methods

The deliberate prevention of pregnancy is called contraception. This can be achieved in several ways. Some contraceptive methods prevent gamete development or releasing gametes. Some contraceptive methods prevent fertilization of gametes. Some methods are available to prevent implantation of an embryo. Unwanted pregnancies are avoided by birth control methods.

Common temporary birth control methods:

- Oral contraceptives for females: Most oral contraceptives contain high concentration of synthetic estrogen and progesterone. High levels of these hormones inhibit GnRH release from hypothalamus and FSH and LH secretions from anterior pituitary through negative feedback. Prevention of LH release blocks ovulation. Inhibition of FSH secretion prevents follicle maturation. Some oral contraceptives contain only high levels of synthetic progesterone (progestin) which thickens cervical mucus so that it blocks entering sperms to the uterus. If fertilization occurs it can interfere with implantation as well.
- Condoms for males: Barrier devices which prevent sperm entry
- IUD (loop) for females: A device placed in the uterus which interferes with fertilization and prevents implantation of a fertilized ovum
- Depo-Provera injection for females: Periodic injection of a synthetic progesterone which thickens cervical mucus and prevents sperm entry. If fertilization occurs it prevents implantation by making the endometrium thin.

Surgical Sterilization (preventive methods of gamete release)

- Vasectomy for males- Prevents release of sperms
- Tubal ligation for females- Prevents ovum from entering uterus

Abortion

- Abortion is the premature termination of a pregnancy in progress.
- Miscarriage: Spontaneous abortion which occurs naturally.
- Induced abortion is intentionally performed (surgical or nonsurgical). Certain drugs can induce abortion non-surgically within the first 7 weeks after conception. They block progesterone receptors in the uterus thereby preventing progesterone from maintaining the pregnancy.

Table 5.6: Sexually transmitted infections

Infection	Pathogen	Main mode of transmission	Main symptoms
Gonorrhea	<i>Neisseria gonorrhoeae</i> bacterium	• Sexual contact, • Mother to child at birth	In males-burning feeling/discomfort when passing urine, yellow discharge with pus from genito-urinary tract.
Syphilis	<i>Treponema pallidum</i> bacterium	• Sexual contact, • Mother to child at birth	Accompanied by fever, headache.
AIDS (acquired immunodeficiency syndrome)	HIV(Human immunodeficiency virus)	• Sexual contact • Transfer of body fluids (blood, serum) • Use of unsterilized needles • Mother to fetus/child :during pregnancy, at birth and during lactation	Sores or painless ulcers on any part of the body (vagina, lips, fingers, nipples), fever, skin rashes Loss of appetite and weight, fever, persistent dry cough, Lymphoma (cancer in lymphatic system), pneumonia and other disease resulting from breakdown of the immune system
Genital herpes	Herpes simplex 2 virus	Sexual contact	Itchy, painful sores around genital area, fever in some cases

Detecting disorders during pregnancy

- Many development problems and genetic disorders can be diagnosed during the gestation period.
- Ultrasound images can be used to analyze the size and condition of the fetus.
- Amniocentesis and chorionic villi sampling –a needle is used to obtain fetal cells from amniotic fluid or tissue surrounding the embryo. Genetic analysis can be done with these samples.
- Newest method is to use a pregnant mother's blood to analyze the genome of the fetus as the mother's blood contains fetal DNA
- But all detectable disorders are untreatable when the embryo is in the uterus and many cannot be corrected even after birth. However, parents can take informed decisions with the help of these tests.

Infertility

Infertility is the inability to conceive offspring. Both men and women can have reproductive defects leading to infertility. The number of couples facing this problem appears to be increasing in the modern society. Certain forms of infertility are treatable.

Modern reproductive technology for resolving infertility problems

- Some infertility problems are resolved by recent scientific and technological advances. This includes hormone therapy, surgery and assisted reproductive technology.
- Hormone therapy– Sometimes, hormone therapy can increase sperm production in the infertile male or egg production in the infertile female.
- Surgery: The ducts in the reproductive system that are formed improperly or have become blocked can be corrected surgically to resolve infertility.
- Assisted reproductive technology

In vitro fertilization (IVF): In vitro fertilization is a series of procedures used to treat infertility problems and assist with the conception of a child. The process involves removal of oocyte(s) from a female ovary and obtaining sperm from a male and combining the oocyte and the sperm to achieve the fertilization under laboratory conditions. The fertilized eggs are incubated until they reach at least 8 cells and then these embryos are transferred to the woman's uterus for implantation and to continue its development. Conventional IVF needs between 50 and 100 thousands of sperm from the male per one oocyte in order to achieve the fertilization. This is due to the fact that in IVF, acrosome reaction has to take place and thousands of sperm cells have to be involved.

Intra-cytoplasmic sperm injection (ICSI): This is also a type of in vitro fertilization method which is used to address male infertility. If mature sperm are defective or low in number, a whole sperm or a spermatid nucleus is injected directly into the cytoplasm of an oocyte that has been removed from the women's ovary. For insemination, ICSI needs only one sperm per oocyte. Unlike in conventional IVF, the sperm which will be inserted into a particular oocyte is already selected in ICSI. The fertilized egg can then be returned to the woman's uterus for implantation.

Support and Movement

The structure and functions of the skeletal systems of animals

In the animal kingdom, three major types of skeletons are found. They are hydrostatic skeletons, exoskeletons and endoskeletons.

Hydrostatic skeleton is a fluid filled body cavity which is enclosed by the body wall. In cnidarians, gastrovascular cavity acts as the hydrostatic skeleton. In some animals such as nematodes and annelids, the fluid filled cavity enclosed by the body wall (e.g., pseudocoelom in Nematoda, coelom in Annelida) consists of two muscle layers (longitudinal and circular muscles) which act antagonistically. The combined effect of muscle contraction and fluid pressure aids in locomotion and maintain the shape and form of the animal. In many animals, the spaces between cells are filled with fluid called interstitial fluid which provides support to these cells.

Exoskeleton is a rigid outer covering of the body of the animal which acts as a skeleton. Different types of exoskeletons are seen in the animal kingdom: Chitinous exoskeleton, calcium carbonate exoskeleton and bony plates. Arthropods possess the exoskeleton which is mainly composed of a non-cellular material, chitin. The chitinous exoskeleton is hardened by proteins or calcium carbonate. Exoskeletons that are made up of calcium carbonate are seen in the molluscs. In some reptiles, bony plates serve as the exoskeleton.

Endoskeleton is a hard skeleton which is buried in the soft tissues of the animal. Different types of endoskeletons are seen in the animal kingdom. These include plates of calcium carbonate (in echinodermates), bones and cartilage (in chordates).

Common functions of the skeletal systems in animals

- Support – All skeletons provide a rigid framework for the body and are resistant to compression and tension forces. They help to maintain the shape of body.
- Protection – The skeleton protects the delicate internal organs.
- Movement – Most skeletons are composed of rigid materials which provide a means of attachment for the muscles of the body. Parts of the skeleton operate as levers on which the muscles can pull. When this occurs, movement takes place.

Functions of the human skeletal systems

- Support
- Protection
- Movement

- Storage and release of calcium under the influence of some hormones (refer competency level 5.7.1)
- Storage and release of phosphates under the influence of some hormones (refer competency level 5.7.1)
- Production of blood cells in the bone marrow

How animals move through water and air?

Swimming: Different groups of animals swim in different ways. Some animals use their legs as oars to push against the water (e.g. insects and four legged vertebrates). Some animals are jet propelled taking water into the body and squirting it out in bursts (e.g. squids). Fishes swim by moving their body and tail from side to side. Aquatic mammals move by undulating their body and tail up and down (e.g. whales and dolphins). Fusiform body shape is a common adaptation for fast swimming animals.

Movement through air: Animals move through air mostly by flying. Gliding downward can occur in some instances. Flying animals use wings to lift the body against the gravity. Wings act as air foil: their shapes alter air currents in a way that helps flying. Fusiform shape of the wings helps to reduce drag force in air.

The human skeleton

Human skeleton is divided into two main parts: axial skeleton and appendicular skeleton.

- Axial skeleton consists of skull, vertebral column, sternum and ribs.
- Appendicular skeleton consists of girdles (pectoral and pelvic) and limb bones.

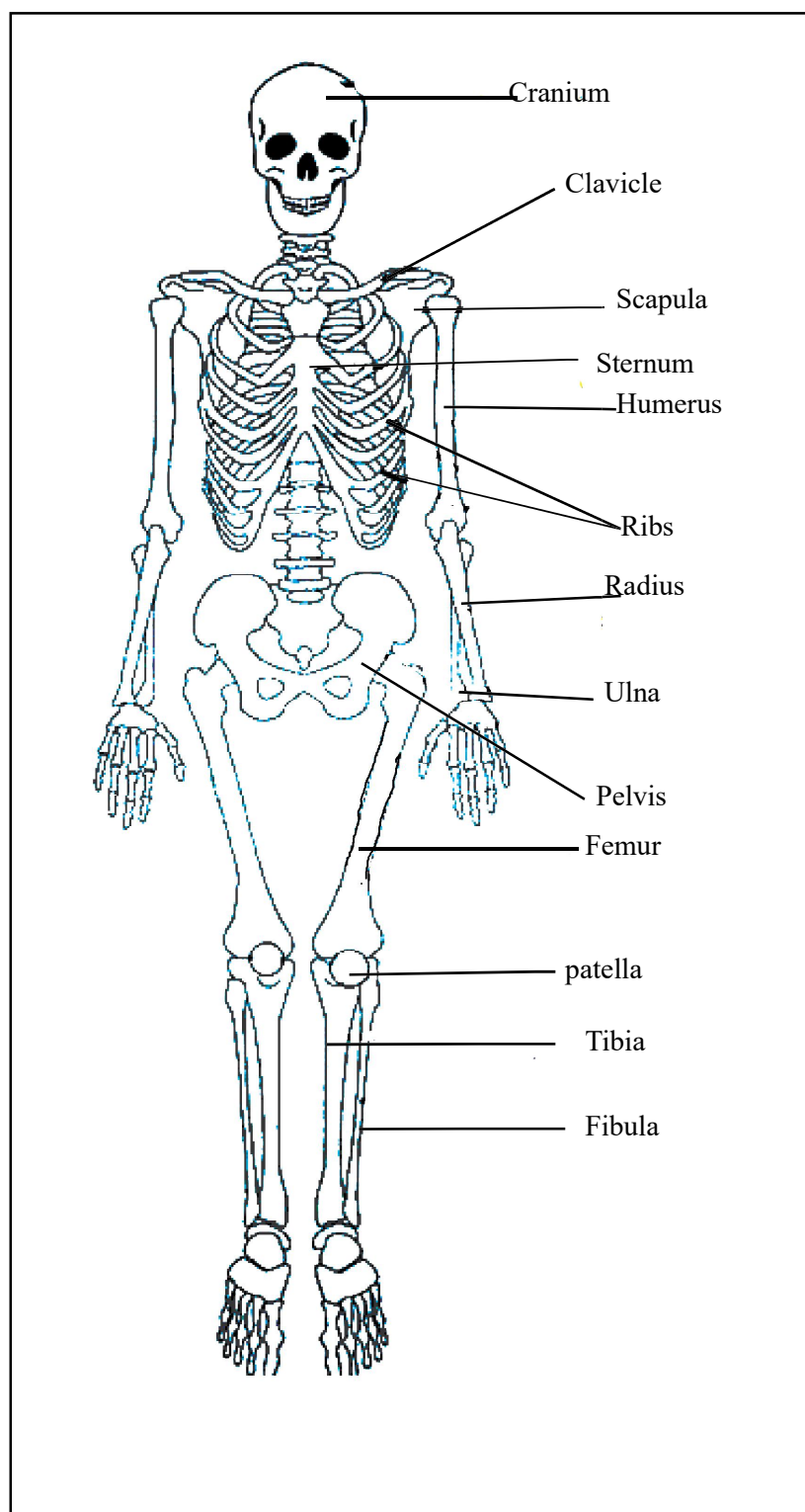


Figure 5.31: The anterior view of the human skeleton

Organization of the human axial skeletal system

Skull

In humans, the skull rests on the upper end of the vertebral column. The human skull consists of 21 bones which are mostly connected together by ossified joints (sutures). Skull is divided into the cranium (brain case) and the face. The bones in the cranium are the frontal bone, two parietal bones, the occipital bone, two temporal bones, the ethmoid bone and the sphenoid bone. In addition to the frontal bone, thirteen other bones form the skeleton of the face. They are two zygomatic bones (cheek), the maxilla (upper jaw bone), two nasal bones, two lacrimal bones, the vomer, two palatine bones, two inferior conchae and the mandible (lower jaw bone).

Human cranial capacity is nearly 1.5 L. Cranium protects and encloses the brain. It also protects the inner ear, middle ear, olfactory organs and eyes. Bony eye sockets provide attachment to the eye muscles that move them. On the inferior surface of the cranium there is foramen magnum to provide passage to spinal cord. Two smooth rounded knobs (Occipital condyles) on either side of the foramen magnum articulates with the first vertebrae (atlas vertebrae) which permits nodding movements.

In the cranium, soft membranous regions called fontanelles are present which allow slight compressions at birth facilitating parturition. Fontanelles become replaced by bones within 1-2 years of life. Immovable joints (sutures) are present between the skull bones to provide more protection. Several air filled cavities lined by ciliated mucous membrane are present in the skull (in the sphenoid, ethmoid, maxillary and frontal bones). They are called sinuses. They all communicate with the nasal cavity. Sinuses provide resonance to voice and reduce the weight of the skull.

Facial region is situated below the cranium. Some facial bones form the walls of the posterior part of the nasal cavity and form the upper part of the air passages. Maxilla and mandible provide ridges in which teeth are embedded. Upper jaw (maxilla) is fused with cranium. Lower jaw (mandible) is movable. Hard palate (bony) and soft palate (cartilaginous) separate the buccal cavity from nasal cavity. Lower jaw articulates with the cranium. Zygomatic arch (formed from parts of zygomatic bone and temporal bone) provides the surface for muscular attachment for moving the lower jaw. Lower jaw (mandible) contains two processes: Condylod process which articulates with the temporal bone to form the temporal-mandibular joint; Coronoid process which gives attachment to muscles and the ligaments. At the base of the skull, occipital condyles (1 pair) are present on the two occipital bones to form a hinge joint with atlas vertebrae. Temporal bone contains three processes: zygomatic process (which forms part of the zygomatic arch), mastoid process and styloid process. They provide surfaces for muscle attachment.

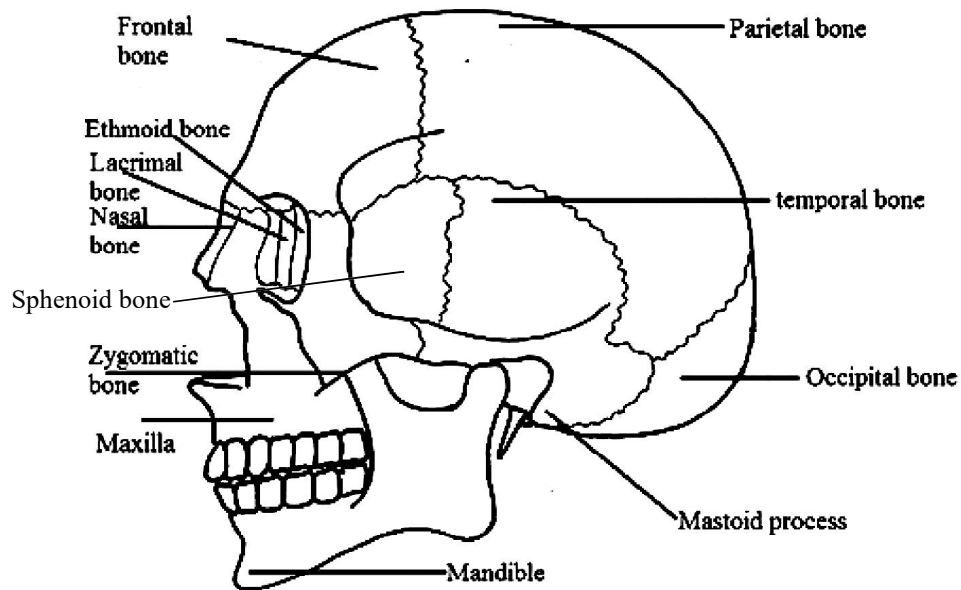


Figure 5.32: The bones of the human skull

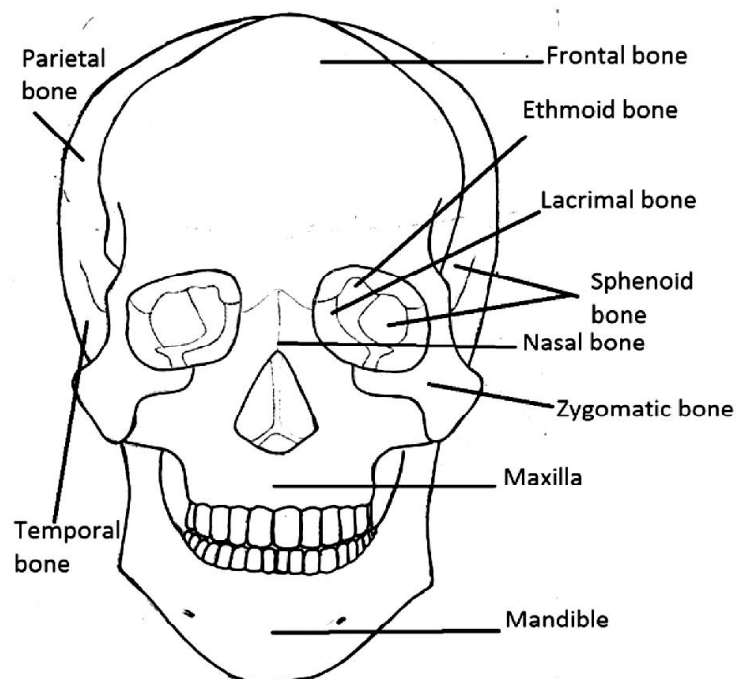


Figure 5.33: The anterior view of the human face (bones in the face)

Vertebral column

Vertebral column is a strong flexible rod consisting of 26 linearly arranged bones. It consists of 24 separate vertebrae extending downwards from the occipital bone of the skull, the sacrum (formed from 5 fused vertebrae) and coccyx (formed from 4 small fused vertebrae). The vertebral column is divided into different regions. There are 4 distinct regions: cervical spine (formed by 7 vertebrae in the neck), thoracic spine (formed by next 12 vertebrae), lumbar spine (formed by next 5 vertebrae), and the sacrum to which the lowest vertebrae of lumbar spine is articulated; the coccyx is situated at the end.

Curvatures of the vertebral column

In humans, there are four curves in the vertebral column: cervical, thoracic, lumbar and sacral. They can be categorized into two main types: two primary curvatures and two secondary curvatures. Main function of the curvatures is the maintenance of the erect posture.

Primary curvatures: In the foetus, there is only one curvature in the vertebral column. When secondary curvatures are formed the primary curvature is retained only in thoracic and sacral regions which are known as primary curvatures. They are concave anteriorly.

Secondary curvatures: Formed after birth, first cervical curvature develops at about 03 months of birth. Then the child can hold his head upright. Second, lumbar curvature develops when the child is around 7-8 months. Then the child can hold his body upright. These secondary curvatures are convex towards the anterior.

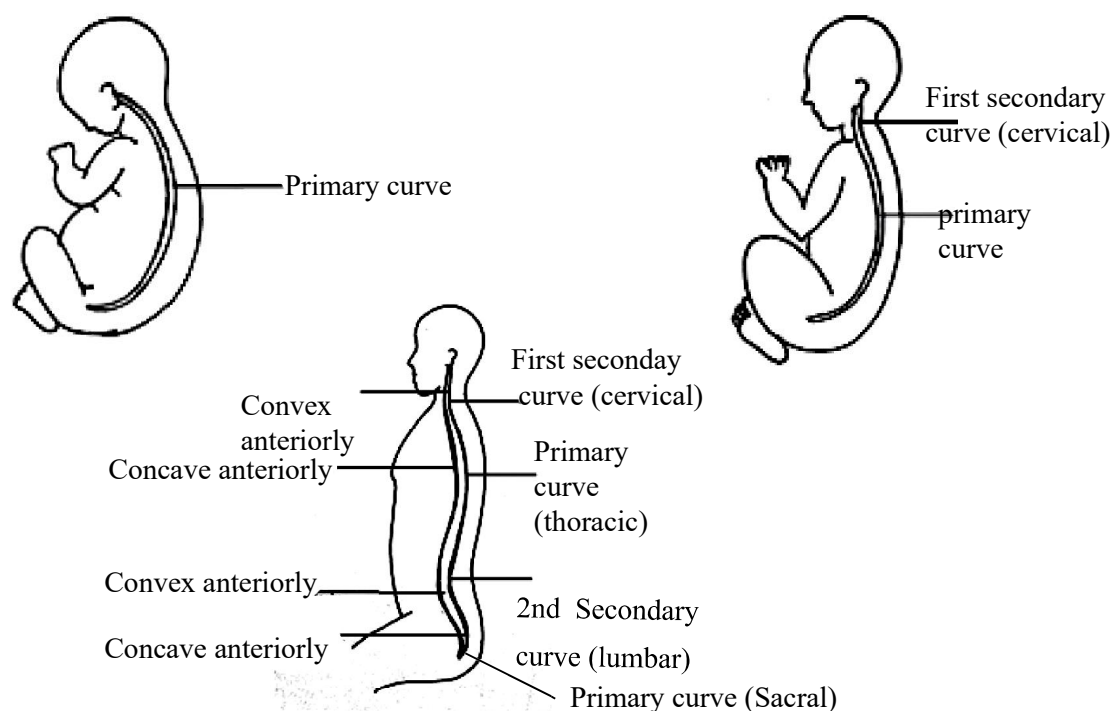


Figure 5.34: The development of curves in the human vertebral column

Types of vertebrae

Structure of a typical vertebra

A lumbar vertebra can be considered a typical vertebra. A typical vertebra consists of the body and the vertebral arch. The body is the largest, broad and flattened part of the vertebrae. The flattened surface of the body of each vertebra articulates with the corresponding surface of the adjacent vertebra so that vertebrae are stacked together in the vertebral column. However the adjacent two vertebrae are not in direct contact with each other as there is a tough pad of cartilage called intervertebral disc between the two vertebrae. The size of the body of the vertebra increases downwards of the vertebral column to support the body weight.

Vertebral arch encloses vertebral foramen which provides passage way for the spinal cord. Processes arising from the neural arch provide surfaces for muscle attachment. Two lateral processes are called transverse processes and the posterior process is called spinous process. The vertebral arch has four articular surfaces: two superior articular surfaces (articulate with the adjacent vertebrae above) and two inferior articular surfaces (articulate with the adjacent vertebrae below).

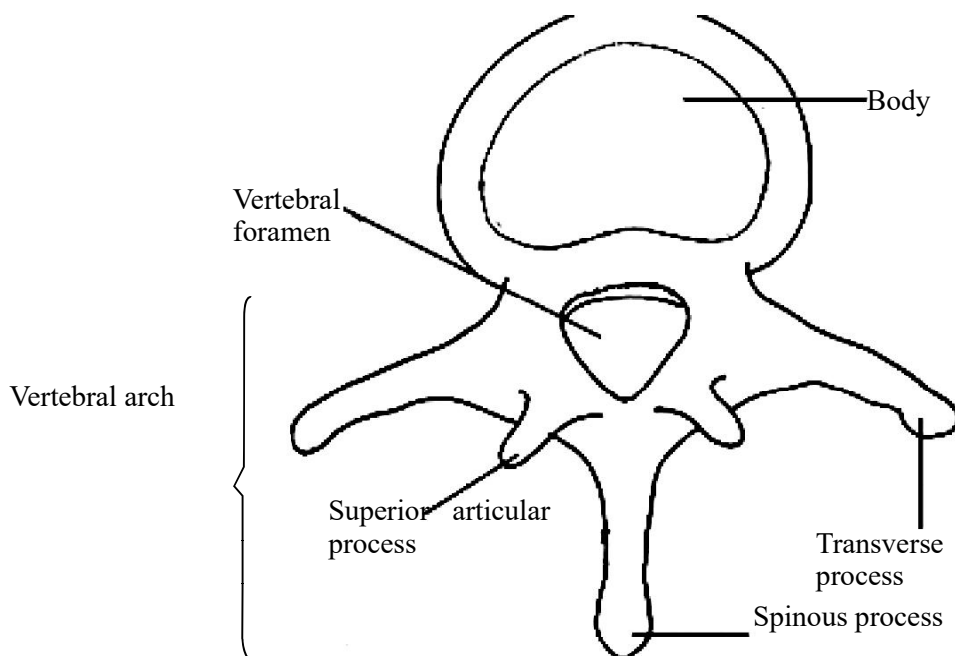


Figure 5.35: The structure of a typical vertebra (lumbar vertebra)

Region specific vertebral characteristics

Cervical vertebrae: Cervical vertebrae are the first seven vertebrae in the vertebral column. When compared to the other types of vertebra, cervical vertebrae are the smallest. Body of the cervical vertebra is smaller compared to the other vertebrae. In addition, transverse processes of a cervical vertebra have a foramen on each side to provide passage for the vertebral artery. The spinous process of these vertebrae is bifid.

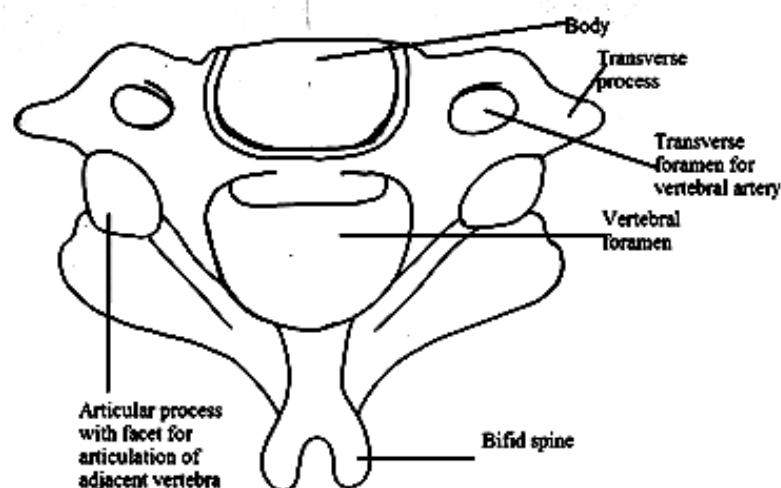


Figure 5.36: The structure of the typical cervical vertebra

The first cervical vertebra is the atlas which is the bone on which the skull rests. It is a ring shaped vertebra with no distinct body or spinous processes. It has two short transverse processes. The atlas contains two flattened facets which articulate with the occipital bone of the skull (condyloid joints), permitting nodding movements. Vertebral foramen of this vertebra is relatively larger to provide the passage of the larger anterior part of the spinal cord.

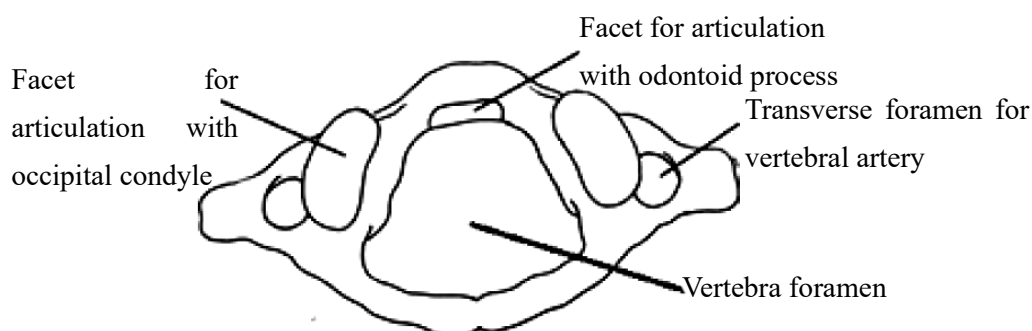


Figure 5.37: The structure of the atlas vertebra

The second cervical vertebra is the axis. It has a small body with a superior projections called odontoid process which articulates with the atlas vertebra above. The head pivots (turns on side to side) on this joint.

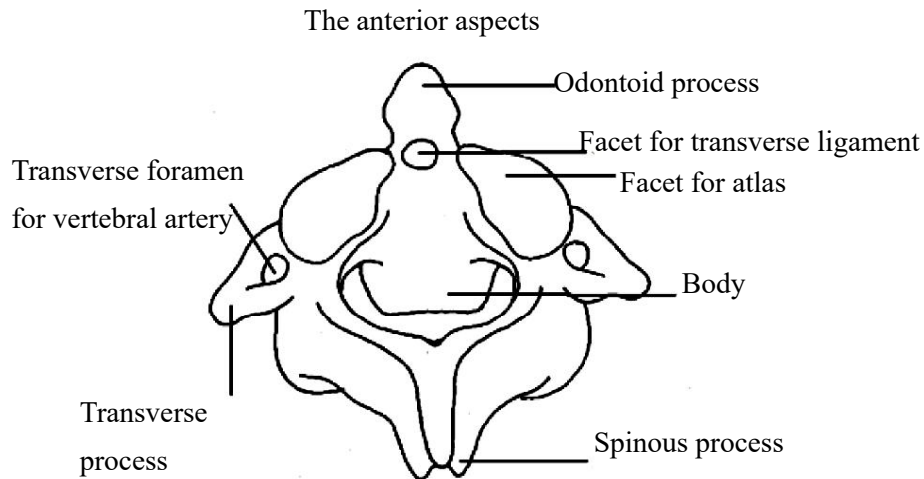


Figure 5.38: The structure of the axis vertebra

Thoracic vertebra: The twelve thoracic vertebrae are larger than cervical vertebrae as this region of the vertebral column has to support more body weight. The body and transverse processes of thoracic vertebrae have facets for articulation with the ribs

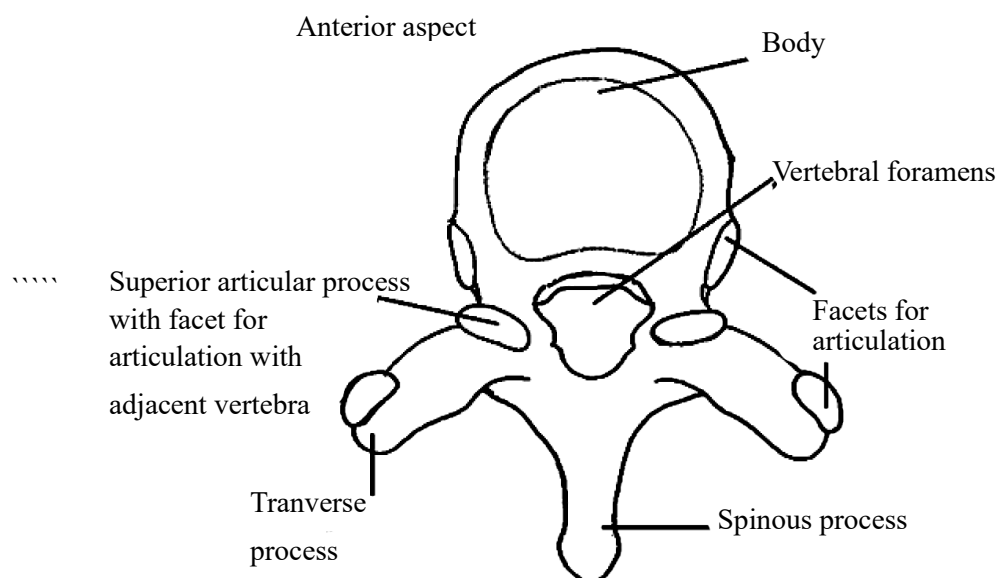


Figure 5.39 : The structure of thoracic vertebra

Lumbar vertebrae: The five lumbar vertebrae are the largest of the vertebrae because they have to support the weight of the upper body. The size of the body of the lumbar vertebrae is larger compared to the other vertebrae. For attachment of the muscles of lower back, the lumbar vertebrae have relatively large spinal processes.

Sacrum and Coccyx: Sacrum is a triangular shaped large bone consisting of five fused rudimentary vertebrae. It has a concave anterior surface. The upper part articulates with the fifth lumbar vertebrae. On each side, sacrum articulates with the pelvic girdle. Inferior tip of the sacrum articulates with coccyx. A series of vertebral foramina are present on each side for passage of nerves. Coccyx consists of fused four terminal vertebrae to form a small triangular bone. The broad base of the coccyx articulates with the tip of the sacrum.

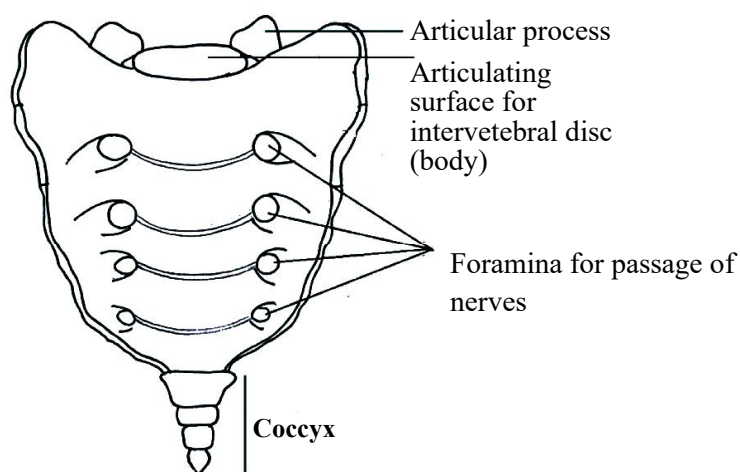


Figure 5.40: The anterior view of the sacrum and coccyx

Common functions of the human vertebral column

The vertebral column helps to maintain the erect posture. It supports the skull and gives attachment to ribs and girdles. It also provides the protection for the spinal cord. Vertebral foramina provide spaces for spinal nerves and blood vessels and lymph vessels. The vertebral column allows flexibility in the body movements. The intervertebral discs act as shock absorbers and protect the spinal cord.

Sternum

Sternum is a long flat bone that forms anterior part of the thoracic cage (which is made up of sternum, ribs and thoracic vertebrae). The uppermost section of the sternum is the manubrium which articulates with the clavicles in the pectoral girdles and the first two pairs of ribs. The body, which is the middle part of the sternum gives attachment to the rest of the ribs. The xiphoid process is the tip of the bone which gives attachment to the diaphragm and muscles of the anterior

abdominal wall. The sternum provides protection to the organs and blood vessels that lie behind it (heart and lungs) from physical damage. The red bone marrow in the sternum is one of the main sites for production of blood cells.

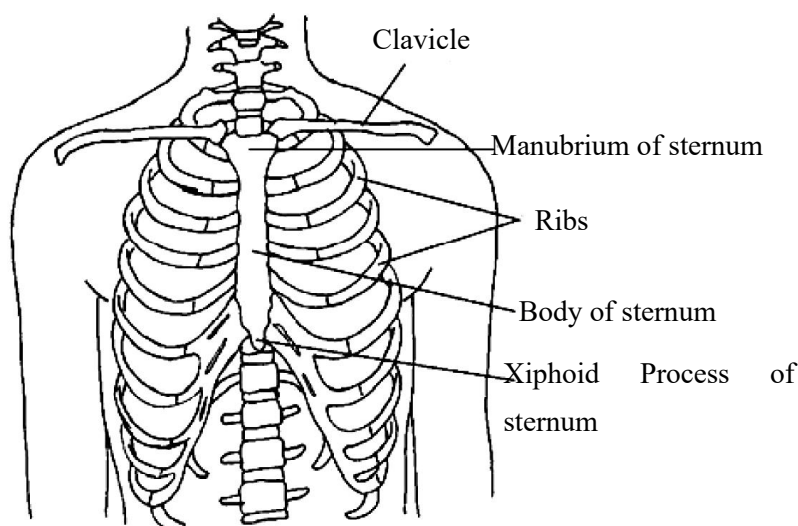


Figure 5.41: The thoracic cage and the location of the sternum

Ribs

The twelve pairs of ribs form the lateral walls of the thoracic cage. They are elongated curved bones. They articulate posteriorly with the thoracic vertebrae of the vertebral column. Anteriorly 7 pairs of ribs articulate with the sternum (true ribs), next 3 pairs articulate with sternum indirectly. In both cases costal cartilages attach the ribs to the sternum. The lowest 2 pairs do not join the sternum (floating ribs). Head of the rib articulates with vertebral bodies, facets of tubercle articulate with transverse process of vertebrae. The thoracic cage which includes the ribs and sternum plays an important role in the mechanism of breathing. Between each ribs intercostal muscles are present which move the rib cage during breathing. The first rib is firmly fixed to the sternum and to the first thoracic vertebra. Therefore it does not move during inspiration. Because it is a fixed point when the intercostal muscle contract they pull the entire rib cage upwards and towards the first ribs. Presence of 12 pairs of ribs and sternum provides protection to the organs such as lungs and heart in the thoracic cavity.

Contribution of human axial skeleton to maintain the upright posture

- Presence of two primary curvatures and two secondary curvatures in the vertebral column. Development of the two secondary curvatures in the vertebral column mainly contribute to maintain the erect posture. (Refer the section on curvatures of vertebral column).

- The size of the vertebrae (especially the body of the vertebrae) become larger towards the end of the vertebral column as they have to support the weight of the upper body (Refer the section on vertebrae).
- The sacral vertebrae are fused to form a triangular shaped large sacrum to support the weight of the vertebral column and internal organs of the body.
- The two occipital condyles (and the foramen magnum) are located inferiorly at the base of the skull close to the center. In the upright position, this arrangement permits proper balancing of the skull on the vertebral column.

The structure and functions of the human appendicular skeleton

Appendicular skeleton

The appendicular skeleton consists of upper limbs with pectoral (shoulder) girdle and lower limbs with the pelvic girdle. Through the pectoral girdle the upper limb forms the joints with the trunk. Pectoral girdle connects upper limb with the axial skeleton. Pectoral girdle consists of two scapulae (shoulder blades) and two clavicles (collar bones). The lower limb forms a joint with the trunk at the pelvic girdle. Pelvic girdle is formed from two hip bones and it is associated with the sacrum.

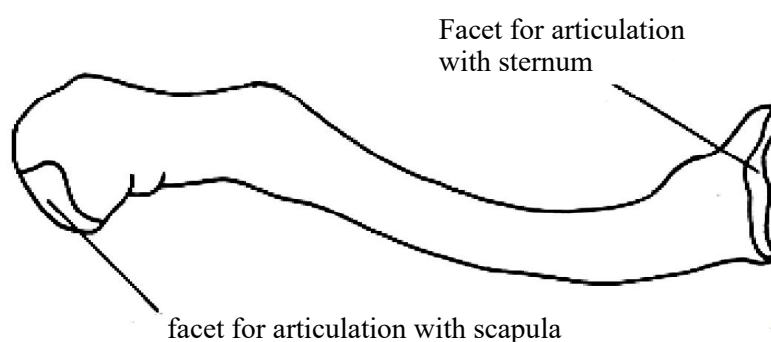


Figure:5.37 -The right clavicle

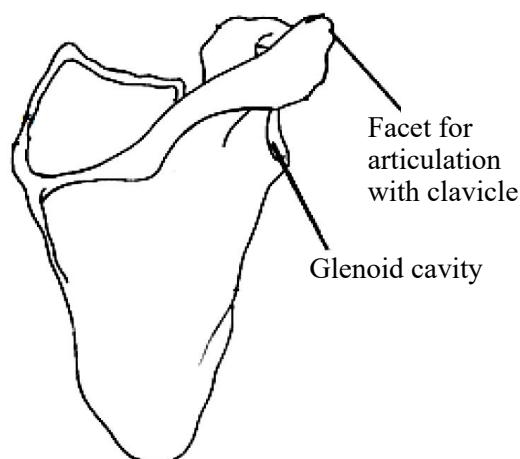


Figure 5.42: The right scapula

Upper limb

Upper limb consists of humerus, radius, ulna, eight carpal bones, five metacarpal bones and fourteen phalanges. Humerus is the bone of the upper arm.

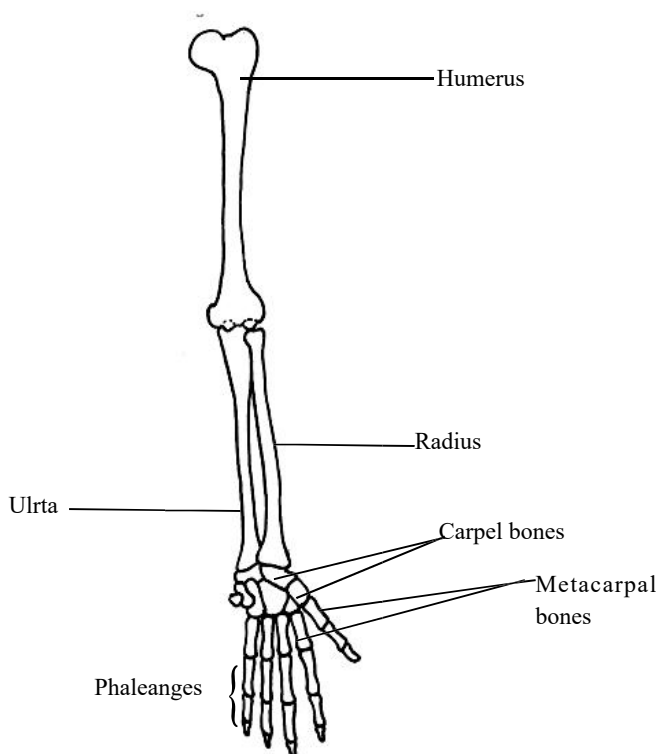


Figure 5.43: Bones of the upper limb

The adaptation of the human upper limb for movement through a wide range

Structure of upper limb is adapted for grasping, weight lifting and movement over a wide range.

Head of the humerus (the bone of the upper arm) forms an incomplete ball and socket joint (shoulder joint) in glenoid cavity of the scapula permitting a vast range of movements. The joint allows for flexion, extension, adduction, abduction, rotation and circumduction.

The distal end of the humerus has two articular surfaces. Through these surfaces, radius and ulna articulate with the humerus at the elbow joint. They articulate with the carpal bones at the wrist joint. Further ulna and radius are articulated with each other at the proximal and distal radio-ulna joints. In addition a fibrous joint connects the bones along their shafts which stabilize their association and maintain their relative position in spite of forces applied from the elbow or wrist. The elbow joint acts as a hinge joint which permits only flexion and extension of the fore arm.

The carpal bones which are arranged in two rows (proximal row and distal row) are closely fitted together so that there is limited amount of movement between them. Proximal row bones are associated with the wrist joint and distal row bones form joints with metacarpal bones. Wrist joint is present between the distal end of radius and three proximal carpal bones. This arrangement allows pronation (palm down) and supination (palm up) of the lower part of the upper limb. In addition the wrist can be flexed, extended, abducted and adducted.

The proximal ends of metacarpal bones in the palm articulate with carpal bones and their distal ends articulate with phalanges. The joints between metacarpal and phalanges allow movement of the fingers and permit the power grip. Fingers may be flexed extended, adducted, abducted and circumducted with the first finger more flexible than the other. The joint present at the base of the thumb between a specific carpal bone and the first metacarpal bone allows more mobility to the thumb than the other fingers. This leads to opposable nature of the thumb which permits the thumb to move perpendicular to the other fingers. This articulation permits precision grip which is unique to man.

Lower limb

Lower limb consists of femur (thigh bone), tibia (shin bone), fibula, patella (knee cap), seven tarsal bones (ankle bones), five metatarsal bones (bones of the foot) and fourteen phalanges (toe bones).

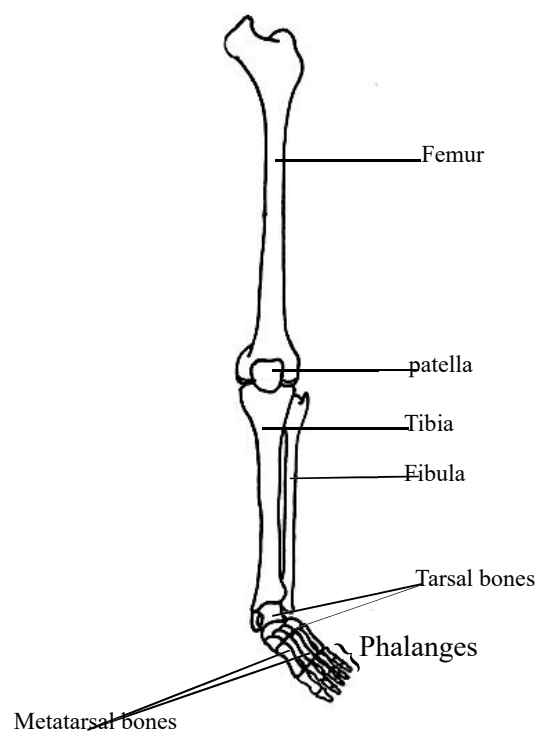


Figure 5.44: Adaptations of the lower limb for the erect posture, bearing of body weight and walking

Structure of the lower limb is adapted for strength, erect body posture, bearing body weight and walking.

Femur is the longest, heaviest and the strongest bone of the body. Head of the femur forms the hip joint (ball and socket joint) with the acetabulum of the hip bone of the pelvis. This hip joint is very sturdy and powerful as it bears all body weight when standing. The lower limb can be extended, flexed, abducted, adducted, rotated and circumducted at the hip joint.

Lower end of femur articulates with tibia and patella to form the knee joint. Tibia is the medial of the two bones. Possible movements at the knee joint are flexion, extension and a rotatory movement that locks the joint when it is fully extended. When this joint is locked it is possible to stand upright for a long period of time.

Femur transmits the weight of the body through the bones below the knee to the foot. All the lower ends of both tibia and fibula articulate with a specific tarsal bone to form the ankle joint. The ankle joint allows rising in tip toe and lifting toes towards the calf.

The arrangement of bones in the foot supported by associated ligaments and muscles gives the sole of the foot an arched or a curved shape. There are two longitudinal arches and one transverse arch in the foot. Curve running heel to toe is called the longitudinal arch and the curve running across the foot is called the transverse arch. In the upright position, these arches of the foot are important in distributing the weight of the body evenly whether stationary or moving.

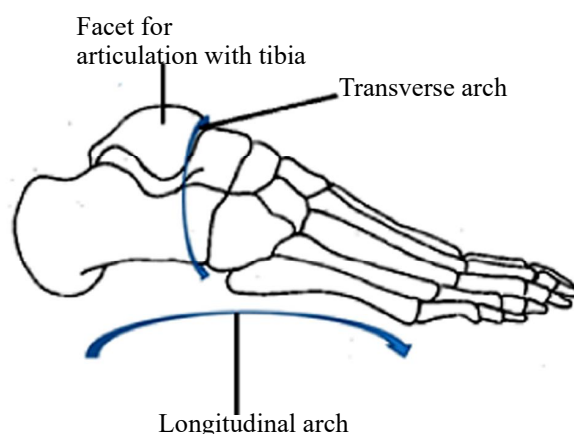


Figure 5.45 :The arches of foot

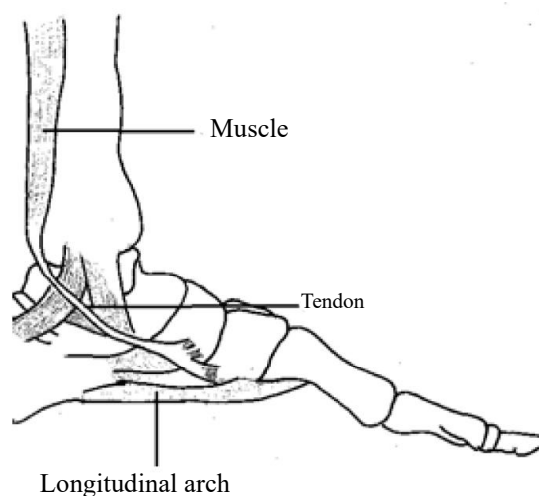


Figure 5.46 :The tendons and ligaments in the foot

Some disorders and abnormalities associated with human skeletal system

Osteoporosis

Osteoporosis is a condition associated with the reduction of bone density due to the exceedance of the bone reabsorption rate over the deposition rate. This gives in fragility to the bone tissue. This condition leads to immobility in joints and may cause fractures, skeletal deformities and bone pain. Factors causing osteoporosis include hormonal imbalances (especially at menopause), calcium deficiency and environmental factors.

Osteoarthritis

Osteoarthritis is a degenerative non-inflammatory disease that causes pain and restricted movements in the affected joints. Articular cartilage at the joints gradually become thinner so that articular surfaces of the bones come in contact and eventually the bones begin to degenerate; the outcome is pain. The cause of osteoarthritis is unknown. But risk factors include excessive use of affected joints, female gender, increasing age, heredity and obesity.

Slipped disc

The bodies of adjacent vertebrae are separated by intervertebral discs which serve as shock absorbers. These intervertebral discs consist of an outer ring of cartilage and a central core of soft gelatinous material. An injury or weakness can cause the inner portion of the intervertebral disc to protrude through the outer ring. This condition is called 'slipped disc'. This leads to pain and discomfort. If the slipped disc compresses a spinal nerve, there can be numbness and pain along the affected nerve. Slipped disc condition can arise when lifting heavy weights without bending knees.

Main types of joints in the human skeletal system

Main types of the joints in the human skeletal system are ball and socket joint, hinge joint and pivot joint.

- **Ball and socket joints**

In these joints, ball shaped head is connected with the cup shaped socket and allows for wide range movements such as flexion, extension, adduction, abduction, rotation and circumduction. There are two ball and socket joints available in the human body: Shoulder joint and hip joint. (Refer upper limb and lower limb)

- **Hinge Joints**

The articulating ends of the bone fit together in such a way so it looks like a hinge of a door. This allows only restricted movements such as flexion and extensions. Examples for hinge joints are elbow joint, knee joint, ankle joint and joints between the phalanges of the fingers and toes. (Refer upper limb and lower limb)

- **Pivot joints**

One bone fits into a hoop shaped ligament that holds it close to another bone and allows it to rotate in the ring formed by the ligament. These joints allow a bone or limb to rotate. For example head rotates by the pivot joint formed by the axis vertebrae within the transverse ligament ring and odontoid process of the atlas. (Refer vertebral column)

Skeletal muscle and mechanism of contraction**Features of skeletal muscle tissue**

The skeletal muscles are generally attached to the skeletal system and mainly cause voluntary body movements. Skeletal muscle tissue is composed of bundles of long cylindrical cells. These cells are aligned parallel to each other along the length of the muscle. Each cell contain multiple nuclei close to the cell membrane. Inside the cell, bundles of myofibrils containing contractile microfilaments are located longitudinally along the length of the cell. Myofibrils in the muscle cell form repeating sections called sarcomeres. The repeating arrangement of sarcomeres within the skeletal muscle cell gives its striated appearance under the microscope. Sarcomeres are the basic contractile units of the striated muscle cell. Like smooth muscle cells and cardiac muscle cells, skeletal muscle cells show excitability or irritability (ability to receive and respond to stimuli), contractility (ability to contract or shorten), extensibility (ability to stretch or contract) and elasticity (ability to return to its original length after being stretched or contracted). The skeletal muscle is under the voluntary control of the somatic nervous system.

Structure of the sarcomere, basic mechanism of skeletal muscle movement

Sarcomeres are the repeating contractile units present within a striated muscle cell. The sarcomere is composed of myofibrils containing contractile thick filaments and thin filaments which are made up of specific proteins. The thin filaments (formed mainly from actin protein) attached at the Z line, a dense stripe which forms the borders of the sarcomere. The thick filaments (formed from myosin protein) are fixed (at the M line) in the middle region of the sarcomere. Sarcomeres are found repeatedly between two Z lines in a skeletal muscle cell. At the resting stage of myofibrils, thick and thin filaments are partially overlapped. At the edge of the sarcomere there are only thin filaments while at the center of the sarcomere only thick filaments are present. Such arrangement of thick and thin filaments in the sarcomeres permits the shortening of the skeletal muscle cell during contraction and return to the original state during relaxation. The mechanical function arising from sarcomeres is produced by actin (found in thin filaments) and myosin (found in thick filaments) proteins.

The skeletal muscle contraction is mainly voluntary and under the control of the somatic nervous system. Upon stimulation, individual muscle cells in the skeletal muscle shortens due to the shortening of its sarcomeres, and thus the whole muscle may contract. Converting muscle contraction to movement needs a skeleton to which the muscles attach. Skeletal muscle contractions pull on the tendons attached to the bones. If contraction of the muscle causes the muscle to shorten, the bone and the body part will move. When the nervous stimulation is stopped, the muscles will return to the original length after being contracted.

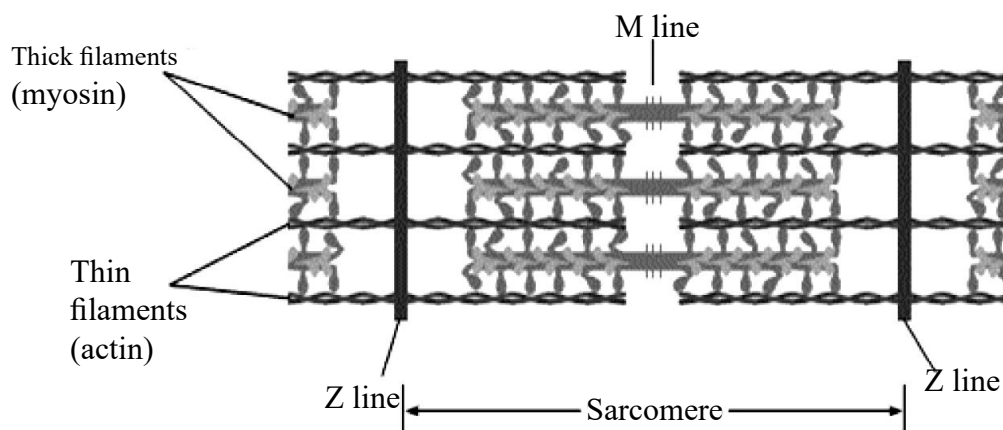


Figure 5.46: The arrangement of a sarcomere

Sliding filament theory is the currently accepted model of striated muscle contraction. According to this theory, when a skeletal (or cardiac) muscle cell contracts, the thick (myosin) filaments and thin (actin) filaments in each sarcomere slide past each other pulling the Z lines at each end of the sarcomere closer to one another shortening the sarcomeres and thus the muscle cell, while the two

groups of filaments in the sarcomere remain at a relatively constant length. Myosin is the motor protein that does muscle contraction by pulling on thin filaments (actin) in muscle cells. Each myosin molecule is composed of 'tail' region and 'head' region. In the thick filaments, these 'tail' regions are bundled together while the 'heads' are sticking out. The thin filaments are composed of actin molecules which have binding sites for the 'head' region of the myosin molecules. The head region of the myosin can also bind with an ATP molecule when its 'low energy state'.

When the ATP molecule is hydrolyzed to form ADP and phosphate while releasing energy, the myosin head enters into the 'higher energy state'. At this state, the myosin head binds to myosin binding site of actin forming a cross bridge. Thereafter the myosin head returns to its lower energy state by releasing ADP and phosphate, which pulls (slides) the thin filament toward the centre of the sarcomere and so shortening the sarcomere. When a new molecule of ATP binds to the myosin head, the cross bridge is broken, myosin head detaches from actin. A new cross bridge cycle begins again. The contraction of muscles requires many number of repeated cycles of binding and releasing. In each cycle, the myosin head is released from the cross bridge and newly bound ATP is hydrolyzed which promotes binding of myosin again to a new actin molecule. This process occurs along the entire length of every myofibril in the muscle cell. Since in the earlier cycle the thin filament has moved towards the centre of the sarcomere, a new binding site for the myosin head region is exposed in the thin filament. The entire process causes the thick and thin filaments in the muscle cell to slide past each other pulling the Z lines at each end of the sarcomere closer to one another shortening the sarcomere.

Many myosin heads can be found in one thick filament. Within one second, each of these heads can form cross bridges. Ca^{2+} and some other proteins also play a major role in muscle contraction. Myosin can only bind to actin when the binding sites on actin are exposed by the action of calcium ions.

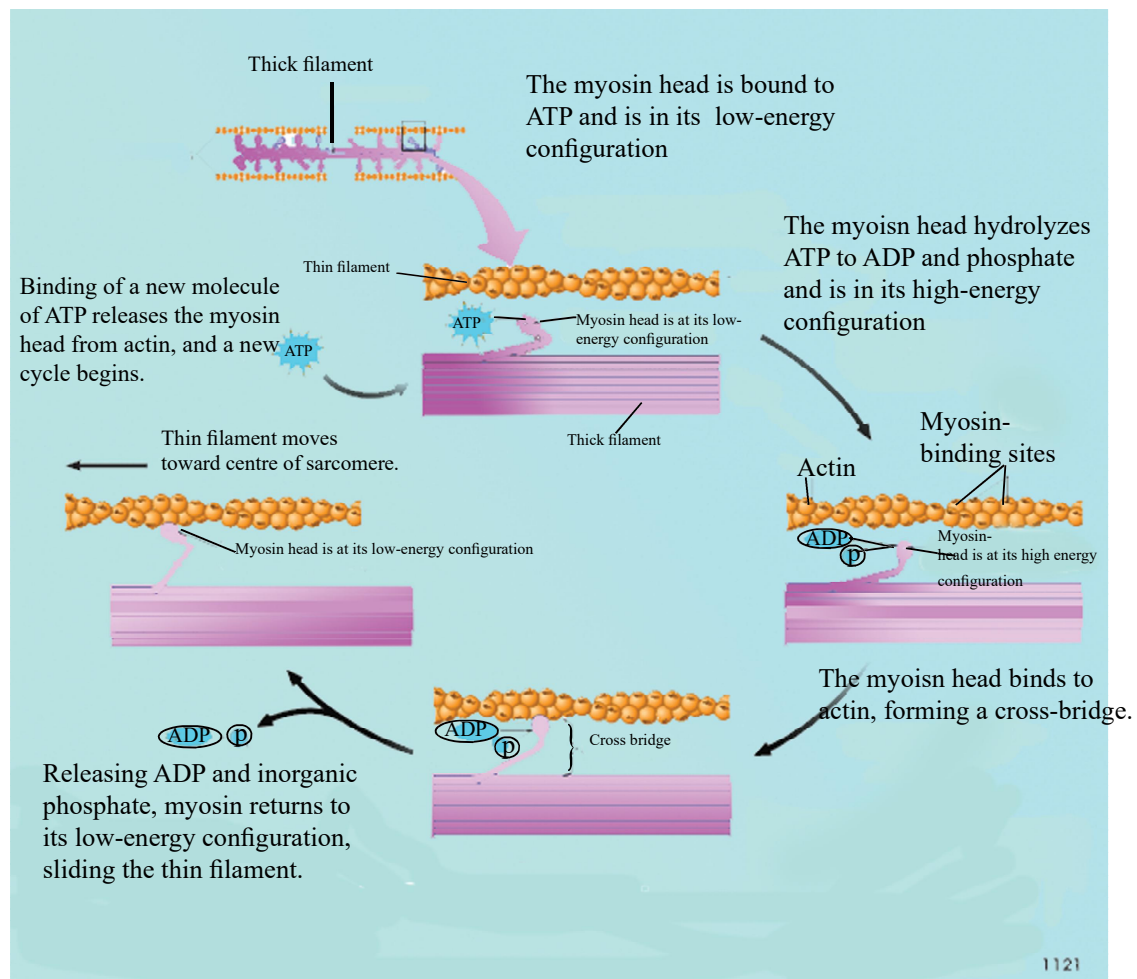


Figure 5.47: Interaction of actin and myosin in skeletal muscle cell contraction

06

Genetics**Scientific basis of Mendel's experiments****Mendelian Heredity (Mendelism)**

Principles in heredity were first formulated by an Austrian Augustinian monk named Gregor Mendel, who is now considered the father of modern genetics. Mendel discovered the basic principles of heredity by breeding garden peas in carefully planned experiments.

His experiments were conducted decades before the concept of chromosomes. The later discovery of chromosomes as the carriers of genetic units supported Mendel's two basic laws of genetics which are now known as **Mendelism**.

Vocabulary in genetics

There are numerous heritable variations among individuals of a population such as brown, green, or blue eyes or black, brown, or blond hair in human population. A heritable feature that varies among individuals of a population, such as hair colour or eye colour is called a **character** in genetics. These heritable variants of a character in an organism, such as brown or blond hair or blue, brown or black eye colour in human are called as **traits**. These traits are transmitted from parents to offspring. Observable traits of an organism is known as phenotype.

Mendel has described about 'heritable factors' in explaining his experimental results. These heritable factors are identified to be **genes** in modern genetics. **Gene** is the basic unit by which genetic information is passed from parent to offspring. It is a DNA sequence residing usually at a specific **locus** on a particular chromosome and contributes to the development of one or more traits by coding for specific proteins or peptides. Locus (Loci in plural) is a fixed position on a chromosome.

There are alternative versions of genes which are called **alleles**. Alleles reside on the same locus of different chromosomes. Alleles vary in their sequence of nucleotides. This change can affect the function of the protein encoded by the gene and thus the phenotype of the organism. Each diploid organism has at least two copies of each gene, residing on the chromosomes received by the two parents. These copies could be identical or could differ from one another. The condition of

having two identical alleles for a given gene is known as **homozygous** state. Alternatively, having two different alleles for a given gene is referred to as **heterozygous** state.

Phenotype is brought about by the interaction between genotype of the individual with its environment. The genetic make up, or set of alleles, of an organism is known as its **genotype**. An individual's genotype could be either homozygous or heterozygous with respect to a given gene.

At heterozygous state, the allele which determines the organism's phenotype by masking the expression of the other is referred to as the **dominant allele**. The trait produced by the dominant allele is known as the **dominant trait**. The allele which does not exhibit any noticeable effect on the organism's phenotype at heterozygous state is referred as the recessive allele. The trait hidden on the recessive allele is the **recessive trait**. However, they express their trait when they exist in homozygous state.

Mendel tracked only those characters that occurred in two distinct, contrasting phenotypic forms, such as tall stem length vs. short stem length or purple flower colour vs white flower colour. Such traits are referred to as **contrasting traits**.

Mendel used only the **pure breeding** (sometimes called **true breeding**) varieties for his experiments. Pure breeding plants are obtained by self-pollinating over many generations, producing only the same variety as the parent plant. These uniform lines produced from self-fertilization of pure breeding varieties over many generations are called **pure lines**.

During his experiments, Mendel cross-pollinated pure-breeding garden pea plant varieties which shows contrasting **traits**. For example, purple-flowered plants were cross bred with white-flowered plants. Mating or crossing of two pure-breeding varieties with contrasting **traits** is called **hybridization**. Parental generation is referred to as the **P generation** (parental generation). Plant progeny resulted from these hybridization events are referred to as **F₁ generation** (First Filial generation, the word filial from the Latin word for "son"). The progeny that results from the self or cross pollination between these F₁ generation plants are known as **F₂ generation** (Second Filial generation).

An organism that is heterozygous with respect to a single gene of interest resulting from a cross between parents having homozygous condition for different alleles of specific gene is referred to as a '**monohybrid**'. Breeding experiment conducted between two organisms with heterozygous condition for a specific character is referred to as '**monohybrid cross**'.

An organism that is heterozygous with respect to two genes of interest resulting from a cross between

parents having homozygous conditions for different alleles of two specific genes is referred to as a '**dihybrid**'. Breeding experiment conducted between two organisms with heterozygous conditions for two specific characters is referred to as a '**dihybrid cross**'.

Breeding an organism having unknown genotype for a specific dominant trait with an organism having homozygous recessive condition for the same specific trait is called a **test cross**. This is usually done to reveal the unknown genotypes for specific dominant traits in an organism.

Monohybrid Cross

Mendel derived his first law of inheritance by following only a single character in one breeding experiment, such as flower colour. He started by crossing pure breeding parents with contrasting **traits**. All the F_1 progeny produced from pure breeding parents are **monohybrids**, meaning that they are heterozygous for the particular character being followed in the cross.

F_1 hybrid pea plants were then self- or cross-pollinated and F_2 generation was produced to explore the traits resulting from a monohybrid cross (Figure 6.1).

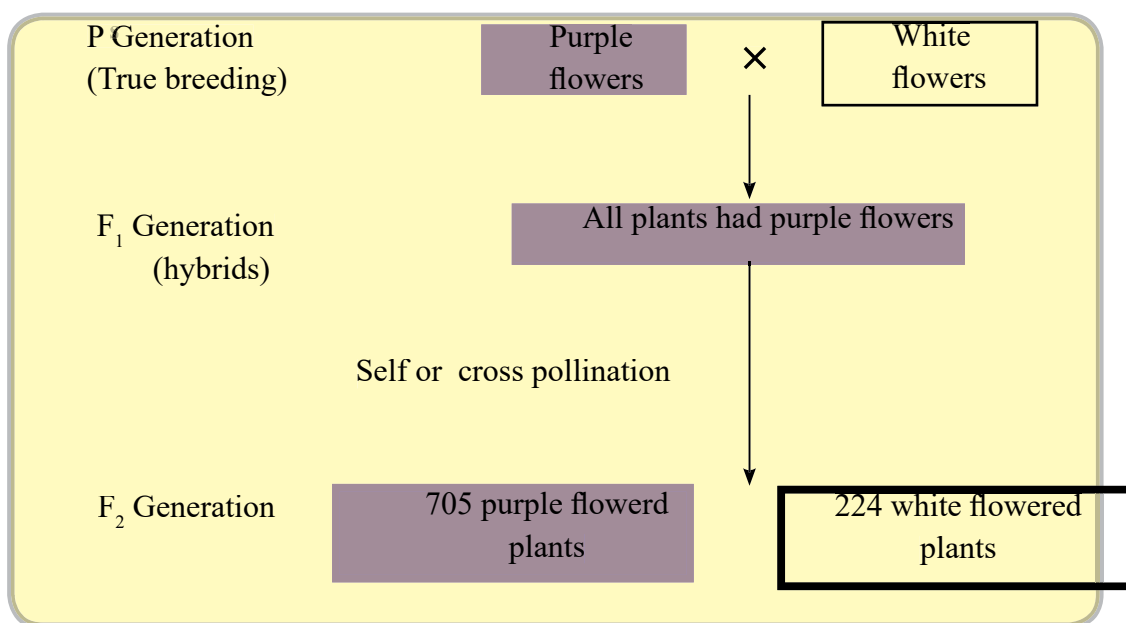


Figure 6.1: Mendel's experiment to investigate the inheritance patterns of a single character over two generations

Mendel, during his experiment, crossed pure breeding purple flowered plants and white flowered plants. Then he allowed the resulting F_1 hybrids to self and cross-pollinate with other F_1 hybrids. Finally, he observed the F_2 generation plants for the colour of the flowers.

During his observation, all F_1 plants produced purple colour flowers. However, in the F_2 generation, both purple and white flowered plants appeared in a ratio of approximately 3: 1.

Among the heterozygote resulted in the F_1 generation, the “heritable factor” responsible for producing white flowers were suppressed in presence of the “heritable factor” that produces purple flowers. As a result, the heterozygotes were all producing purple colour flowers. Therefore, the ‘heritable factor’ for purple flower colour is dominant to the white (dominant trait). Accordingly, the factor for white flower colour is referred as the recessive trait.

Mendel observed that the same pattern of inheritance consistently occurred in six other characters; position of the flower, colour of the seed, shape of the seed, shape of the pod, colour of the pod and the length of the stem.

Mendel’s first law of inheritance: The law of segregation

Mendel’s first law was put forward to explain the 3:1 inheritance pattern observed among the F_2 offspring in his monohybrid experiments using Pea plants.

As per his hypothesis, each ‘heritable character’ is determined by two “heritable factors” which are known as alleles. During the formation of gametes, the alleles for a ‘heritable character’ are separated and get in to each of the gametes formed. This is now known as **Mendel’s law of segregation or Mendel’s first law in inheritance**.

Analyzing genotype and phenotype ratios using Punnett square

In Pea plants, Mendel observed two different **traits** based on stem lengths; tall and dwarf. For his experiments, pure breeding tall and dwarf Pea plants were selected for cross pollination. Thereafter the F_1 generation was self pollinated in order to obtain F_2 generation.

During self pollination of F_1 hybrids, gametes carrying different alleles fuse randomly. Such random fusion of gametes produces zygotes with four genetic combinations. A Punnett square can be used to illustrate these genetic combinations. A Punnett square is a graphical representation of the possible genotypes of an offspring arising from a particular cross or breeding event. An example is given figure 6.2.

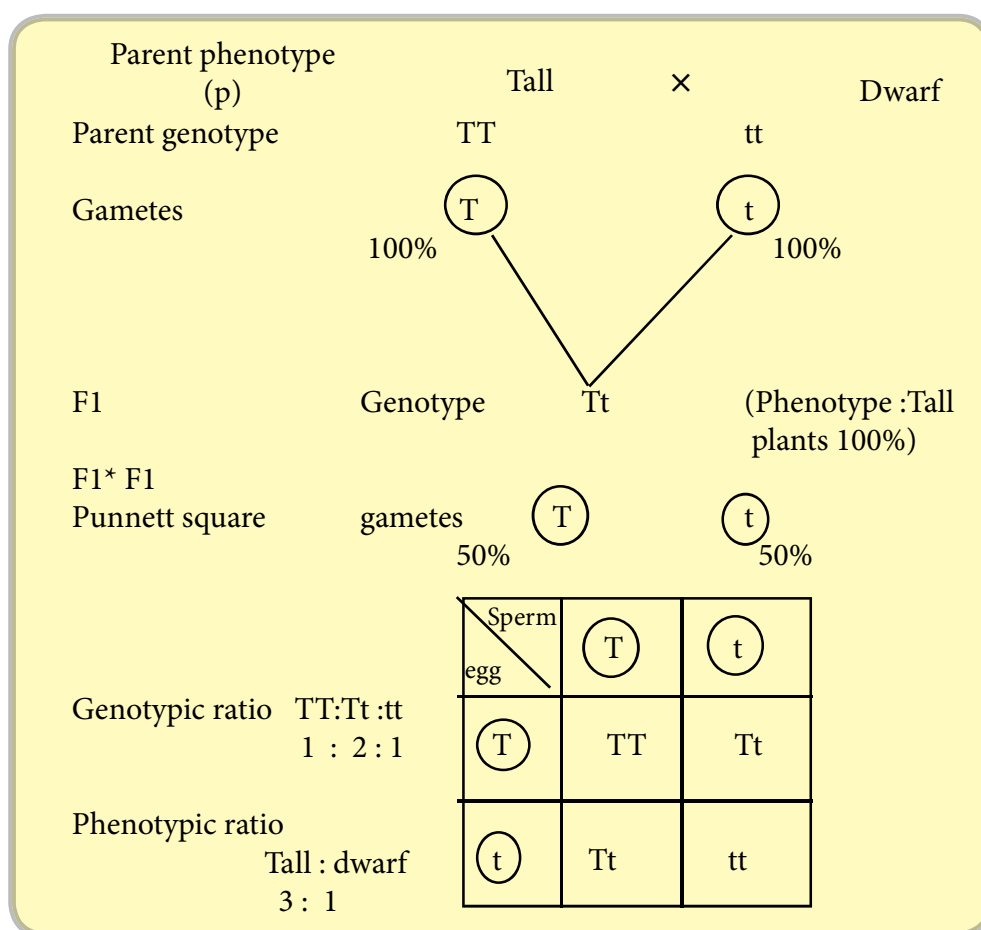


Figure 6.2: Mendel's law of segregation shown with the aid of the punnett square

Dihybrid Cross

Mendel identified his second law of inheritance by following two characters at the same time, using **dihybrid crosses**. A cross between two homozygous (true breeding) organisms with contrasting traits for two specific characters being followed is known as a **dihybrid cross**. The aim of the Mendel's dihybrid cross experiment was to find out whether the alleles for one character assort into gametes dependently or independently of the alleles of the other character.

Mendel crossed a true-breeding plant with yellow-round seeds with a true breeding plant with green-wrinkled seeds (Figure 6.3). The cross produced dihybrid F_1 plants, all of which have yellow-round seeds. As shown by the monohybrid crosses, the allele for yellow seeds is dominant (Y) over the allele for green seeds (y) which is recessive. likewise, the allele for round seed is dominant (R), and the allele for wrinkled seed is recessive (r). The F_1 hybrids, are heterozygous for the two characters being followed in the cross (YyRr). The cross between F_1 dihybrids produced the F_2 generation.

This lead towards two alternative hypothesis for inheritance which predict different phenotypic ratios as shown in **Figure 6.3**.

1. The two characters could be transmitted from parents to offspring as a package. The dominant Y and R alleles or the recessive y and r alleles are passed together, generation after generation. This is called **dependent assortment** of alleles.

According to this hypothesis only two types of gametes are possible; i.e. YR and yr. Thus the phenotypic ratio of the F₂ generation would be similar to that of a monohybrid cross (3:1)

2. The two characters (seed colour and seed shape) could be transmitted from parents to offspring independent of each other i.e. Y allele could be passed either with R or r allele vice versa. This is called **independent assortment** of alleles.

This hypothesis predicts four different allelic combinations for a bi-allelic locus and thus four different types of gametes from F₁ generation; i.e. YR, Yr, yR and yr. According to this, both male and female gametes have four possibilities for each. Therefore, during the union of male and female gametes, there are 16 (4 × 4) equally probable ways in which the alleles can combine to produce the F₂ generation. As shown in Figure 3, these combinations would give rise to four different phenotypes with a ratio of 9:3:3:1 (nine Yellow round to three Green-round to three Yellow-wrinkled to one Green-wrinkled).

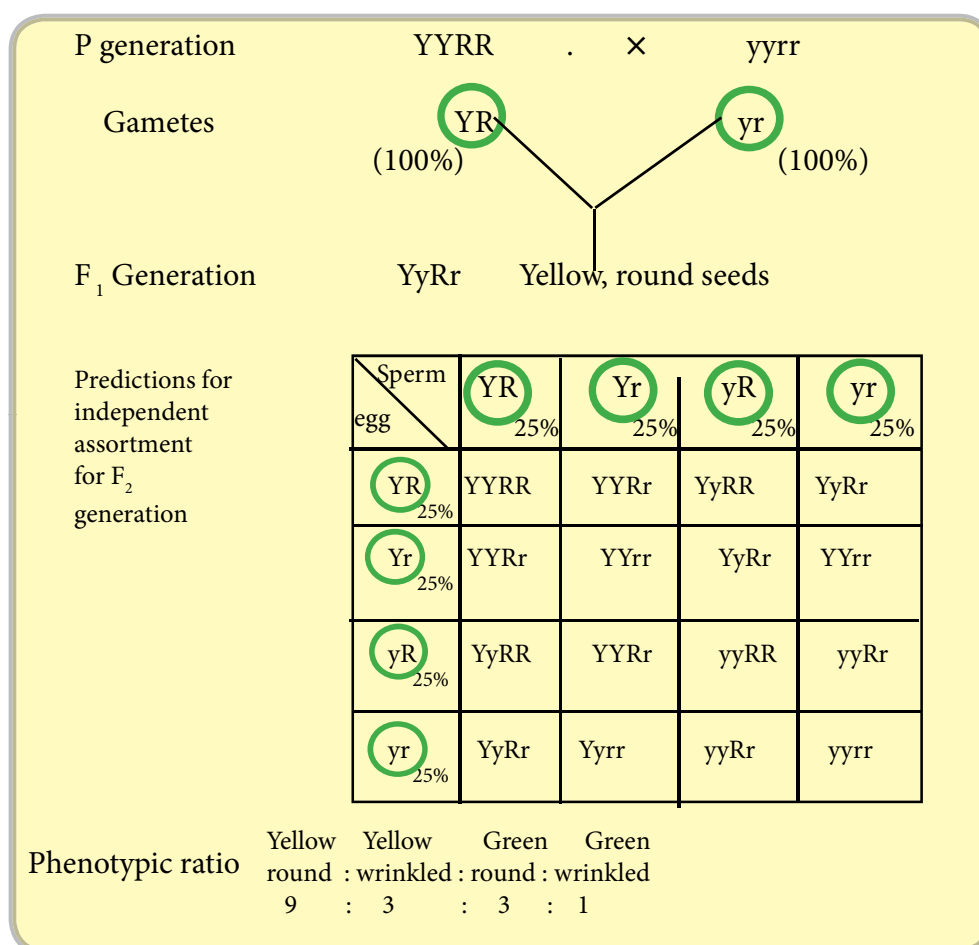


Figure 6.3: Alternative forms of inheritance patterns possible in a dihybrid cross

Mendel's experiment produced the four distinct phenotypes (Yellow-round, Green-round, Yellow-wrinkled and Green-wrinkled seeds) predicted in the second alternative hypothesis in the ratio of 9:3:3:1. This showed that, the alleles responsible for each trait assorted independently of those of the other.

Mendel's second law of inheritance (The law of independent assortment)

Based on these experiments, Mendel put forward his **second law of inheritance - the law of independent assortment**. The law states that, alleles separate and pair up independently during the formation of gametes. As a result of that, two or more genes assort independently irrespective of the other.

However, according to current knowledge, this condition applies to two circumstances only;

- To genes located on different chromosomes (genes on non-homologous chromosomes)
- To genes located far apart on the same chromosome

Success of Mendel's experiments

Mendel followed a scientific approach in conducting his experiments. The following features in his experiments helped him to unravel the two fundamental principles of heredity.

- Mendel carried out thousands of genetic crosses of any given kind. This allowed his results to closely resemble the probability predictions. Usually, the larger the sample size, the closer the results to the value predicted based on the probability.
- He kept accurate records of his results. These records helped him to trace the patterns which otherwise would go unnoticed.
- He usually followed up each cross for at least two offspring generations (F_1 and F_2). This helped him to uncover some of the traits hidden in the F_1 generation.
- He did a quantitative analysis of the phenotypes of the resulting offspring.

Desirable properties in garden peas for genetic experiments

Garden peas (*Pisum sativum*) carry following desirable properties which makes it a suitable organism to study patterns of inheritance.

- Pea plants are available in many varieties with contrasting traits.
- The generation time is short.
- A large number of offspring is produced from each cross.
- Crossing between the plants could be strictly controlled (self/ cross pollination).

Probability laws and Mendelian inheritance

Mendel's laws of segregation and independent assortment reflect the same rules of probability that applies to tossing coins, rolling dice, and drawing cards from a deck. **Probability** measures

how likely an event is to occur out of the number of possible outcomes. It is calculated by dividing the number of events of interest by the number of total possible outcomes.

1. The probability scale ranges from 0 to 1.

An event that is certain to occur has a probability of 1, while an event that is certain not to occur has a probability of 0.

During allele segregation in a F_1 plant (heterozygous) of a monohybrid cross,

-probability of each egg carrying the dominant allele = $1/2$

-probability of each egg carrying the recessive allele = $1/2$

2. The probabilities of all possible outcomes for an event add up to 1.

During allele segregation in a heterozygous F_1 plant, probability of all events (having dominant and recessive alleles) = $1/2 + 1/2 = 1$

3. When the occurrence of an event does not affect the occurrence of another event (independent events), the probability of simultaneous occurrence of both events can be obtained by multiplying the probability of one event by the probability of the other event. This is known as the **Multiplication Rule** or **Product rule in Probability**.

In Mendel's monohybrid crosses, for a F_2 plant to have wrinkled seeds (rr), both the egg and the sperm that come together must carry the r allele.

The probability that the egg will have an r = $1/2$

The probability that the sperm will have an r = $1/2$

The probability of both gametes at fertilization carrying r allele = $1/2 \times 1/2 = 1/4$

4. The probability that any one of two or more mutually exclusive events will occur is calculated by adding their individual probabilities. This is the addition rule or sum rule of probability.

There are two possible mutually exclusive ways for producing F_2 heterozygotes.

i. *The dominant allele come from the egg and the recessive allele from the sperm; the probability of the event = $1/4$ (according to example in above 3rd sentence)*

ii. *The recessive allele from the egg and the dominant allele from the sperm; the probability of the event = $1/4$ (according to example in above 3rd sentence)*

So, the probability of getting an F_2 heterozygote = $1/4 + 1/4 = 1/2$

Prediction of the inheritance patterns in multifactorial crosses

When the pattern of inheritance of two or more characters of an organism is being traced during a genetic cross, it could be called as multifactorial cross. Finding out the outcomes of a multifactorial cross through a Punnett square may be a difficult task. Therefore, applying rules of probability may be useful to predict the outcomes of a multifactorial crossing.

According to the law of segregation, a multifactorial cross can be considered equivalent to multiple independent monohybrid crosses occurring simultaneously.

E.g. 1: Dihybrid cross for seed color and seed shape,

Probabilities for seed color (based on the Punnett square for monohybrid cross)

Seed color	
Genotype	Probability
BB	$\frac{1}{4}$
Bb	$\frac{1}{2}$
bb	$\frac{1}{4}$

Seed shape	
Genotype	Probability
RR	$\frac{1}{4}$
Rr	$\frac{1}{2}$
rr	$\frac{1}{4}$

B: dominant allele for black colour seed b: recessive allele for brown coloured seeds

R: dominant allele for round shaped seeds r: recessive allele for wrinkled shaped seeds.

The probability of each of the genotypes in the F_2 generation can be determined by using the multiplication rule.

Probability of BbRr = $\frac{1}{2}$ (probability of Bb) $\times$ $\frac{1}{2}$ (probability of Rr) = $\frac{1}{4}$

Probability of bbRr = $\frac{1}{4}$ (bb) $\times$ $\frac{1}{2}$ (Rr) = $\frac{1}{8}$

Probability of bbrr = $\frac{1}{4}$ (bb) $\times$ $\frac{1}{4}$ (rr) = $\frac{1}{16}$

E.g. 2: A trihybrid cross for flower color, seed color and seed shape.

Y: Dominant allele for yellow colour petals y: Recessive allele for white coloured petals

B: Dominant allele for black coloured seeds b: Recessive allele for brown coloured seeds

R: Dominant allele for round seeds r: Recessive allele for wrinkled alleles

YyBbRr $\times$ yyBbrr

(YyBbRr): Yellow coloured petals with black round seeds

(yyBbrr): white coloured petals with black round seeds seeds

Flower color	
Genotype	Probability
YY	0
Yy	$\frac{1}{2}$
yy	$\frac{1}{2}$

Seed shape	
Genotype	Probability
BB	$\frac{1}{4}$
Bb	$\frac{1}{2}$
bb	$\frac{1}{4}$

Shape of the seeds	
Genotype	Probability
RR	0
Rr	$\frac{1}{2}$
rr	$\frac{1}{2}$

Assume that unknown pea plant is having genotype of TT

The above crossing has resulted 640 plants in the F1 generation. Determine the number of plants exhibits dominant phenotype for at least two characters.

1. Possible genotypes with their individual probabilities for above conditions

$$YyBBRr: \frac{1}{2} (\text{probability of } Yy) \times \frac{1}{4} (BB) \times \frac{1}{2} (Rr) = 1/16$$

$$YyBbRr: \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = 1/8$$

$$YyBBrr: \frac{1}{2} \times \frac{1}{4} \times \frac{1}{2} = 1/16$$

$$YyBbrr: \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = 1/8$$

$$YybbRr: \frac{1}{2} \times \frac{1}{4} \times \frac{1}{2} = 1/16$$

$$yyBbRr: \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = 1/8$$

$$yyBBRr: \frac{1}{2} \times \frac{1}{4} \times \frac{1}{2} = 1/16$$

2. Probability of exhibiting at least two dominant characters = $\frac{1}{16} + \frac{1}{8} + \frac{1}{16} + \frac{1}{8} + \frac{1}{16} + \frac{1}{8} + \frac{1}{16} = \frac{10}{16} = \frac{5}{8}$

3. Number of plants expected to exhibit at least two dominant characters = $\frac{5}{8} \times 640 = 400$ plants

The testcross

This is a deliberate breeding process performed in order to determine the unknown genotypes. The genotype of an individual showing dominant trait may be due either to double dominant genotype or to heterozygous status. This involves the crossing of an organism having unknown genotype for a selected phenotype along with another organism from same species having homozygous recessive condition for same character.

Testcross performed during the monohybrid cross is called as monohybrid testcross. On the other hand, test cross performed during the dihybrid cross is called as dihybrid testcross.

Monohybrid testcross

Let's consider an example for monohybrid testcross. In this example we want to know the genotype of the given tall pea plants. In order to do that, we will cross the tall pea plant with the dwarf pea plant. Since dwarf is a recessive trait the genotype of it will be tt.

There may be two possible genotypes for tall pea plants;

1. TT
2. Tt

Assume that unknown pea plant is having genotype of TT

Cross between TT and tt will result the following.

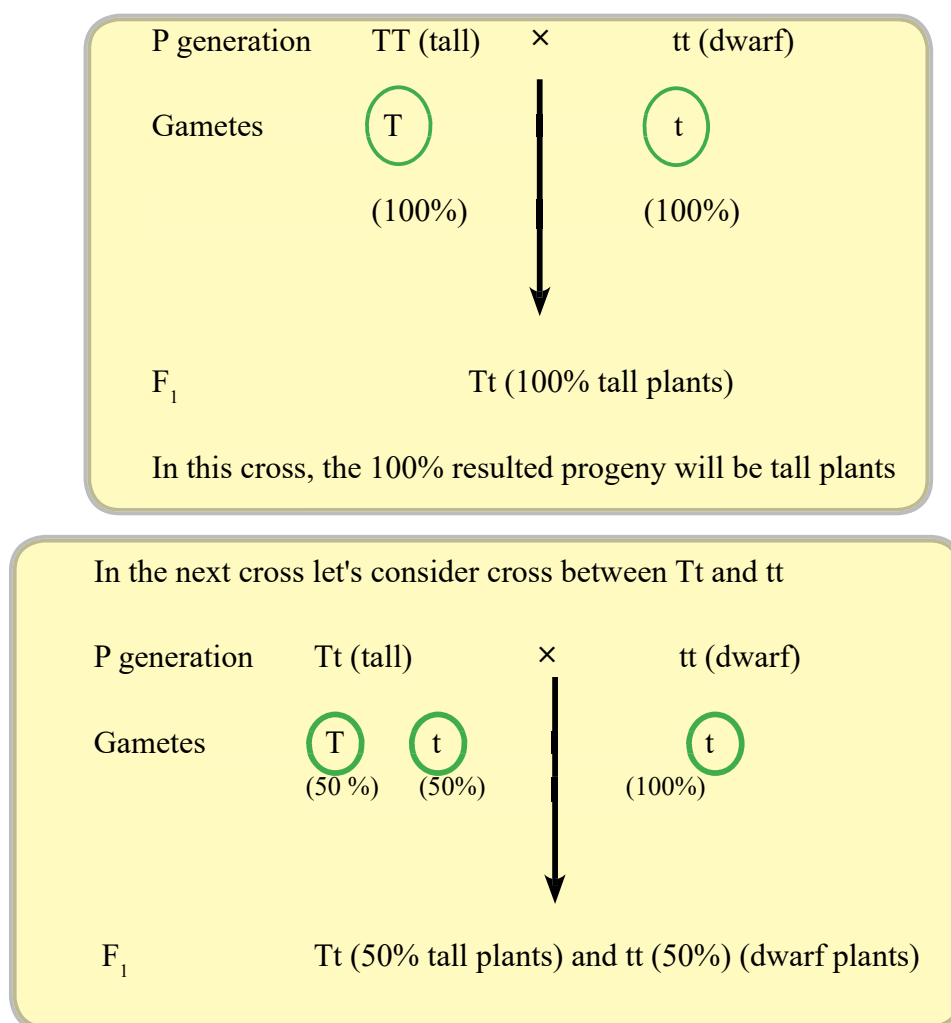


Figure 6.4: Two possible outcomes of a monohybrid test cross

Dihybrid testcross

When an individual with dominant traits for two characters (e.g. RrBb) is crossed with a pure recessive for both characters (wwbb), it is known as a **dihybrid testcross**.

For example, let's consider a cross between plants having red coloured flower petals with black coloured seeds and white coloured flower petals with brown coloured seeds.

The possible genotypes for red coloured petals with black coloured seeds are RrBb/ RRBb/RrBB/ RRBB and for white coloured petals with brown coloured seeds is wwbb.

If the unknown phenotype is RrBb;

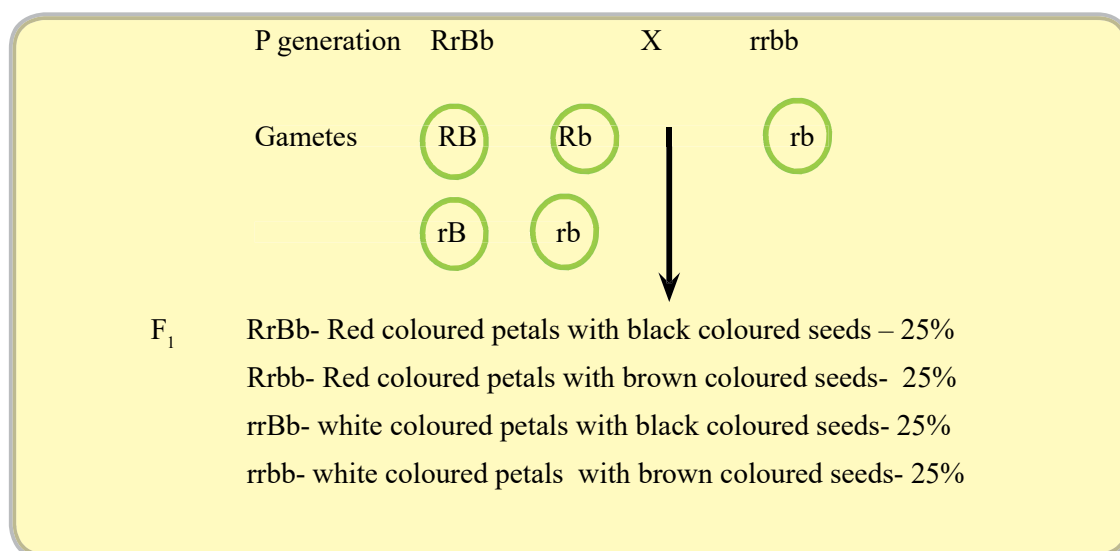


Figure 6.5: (a) Possible outcomes of a dihybrid test cross

In this example, above four phenotypes are possible with 1:1:1:1 ratio.

If the unknown phenotype has the RRBb genotype, the resulting dihybrid cross will produce following phenotypes;

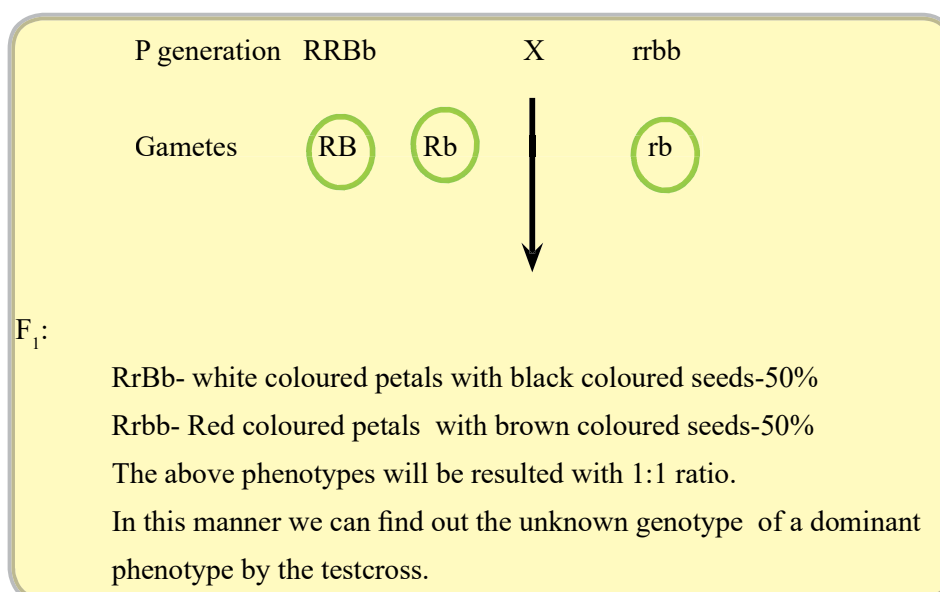


Figure 6.5: (b) Possible outcomes of a dihybrid test cross

Patterns of inheritance of Mendelian characteristics in humans

Common Mendelian characters

Many human traits follow Mendelian patterns of inheritance. Some common examples are stated below;

Attached or detached earlobe:

The extent to which the earlobe is attached to the head is inherited in the Mendelian pattern. The attached earlobe is a recessive trait. Presence of both copies of the recessive allele (homozygous recessive condition) for ear lobe attachment would result in attached ear lobe.

Widow's peak:

The pointed contour of the hairline on the forehead is known as Widow's peak. It is due to a dominant allele, W. Therefore, all individuals who lack a widow's peak must be homozygous recessive (ww).

Dimples on cheek:

Cheek dimples are a genetically transmitted trait found in the muscle of the cheek.

When a person smiles, the shorter muscle on the face pulls up the facial skin. This, in turn, creates a slight depression in the skin, which is called dimple. Dimples often occur on both the cheeks. A single dimple on one cheek is a rare phenomenon. Dimple is a dominant trait and inherited in Mendelian fashion.

Bent thumb (Hitchhiker's thumb) and Straight thumb

Hitchhiker's thumbs is a condition where thumb bend backwards while stretching due to the hyper extensibility of interphalangeal joints. Having the dominant 'S' allele would produce the dominant phenotype of straight thumb. The absence of the dominant alleles would allow the thumb to bend.

Rolling or non-rolling tongue

The ability to roll the lateral edges of the tongue upwards into a tube is known as tongue rolling. The tongue's intrinsic muscles allow some people to form their tongues into specific shapes. Rolling the tongue into a tube shape is a dominant trait with simple Mendelian inheritance.

Pedigree analysis

Diagrammatic representation of the inheritance of a particular trait within a given family tree, is called pedigree chart. It is constructed by collecting data for many generations within a given family, so that the pattern of inheritance can be understood.

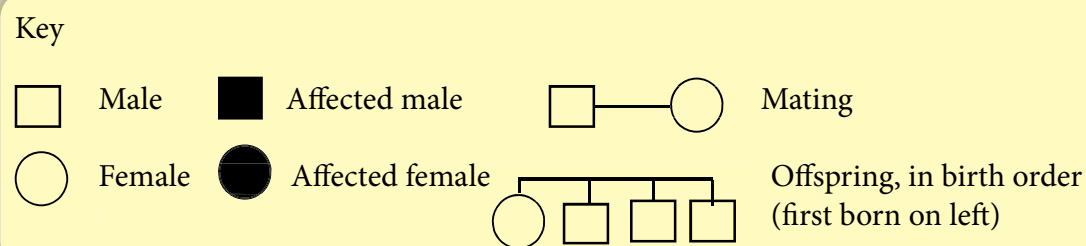


Figure 6.6: Standard Pedigree Symbols

Common Mendelian characteristics in humans analysed with pedigree charts

Widow's peak:

The inheritance of the trait, widow's peak, over three generations in a particular family is represented in the pedigree chart given below. As shown in figure 6.7, only one grand parent had widow's peak, out of the two pairs. Since widow's peak is a dominant character, grand parents without widow's peak should be homozygous recessive (ww) for the trait. In the next generation some individuals showed the widow's peak, while the others did not. The two grand parents, who express the widow's peak should be heterozygous (Ww) for the condition. Likewise, the two parents of the third generation, who are showing widow's peak should be heterozygous for it, as one of both their parents (1st generation) are homozygous recessive (ww). The third generation with widow's peak can carry either WW or Ww genotype, as both their parents are having the trait.

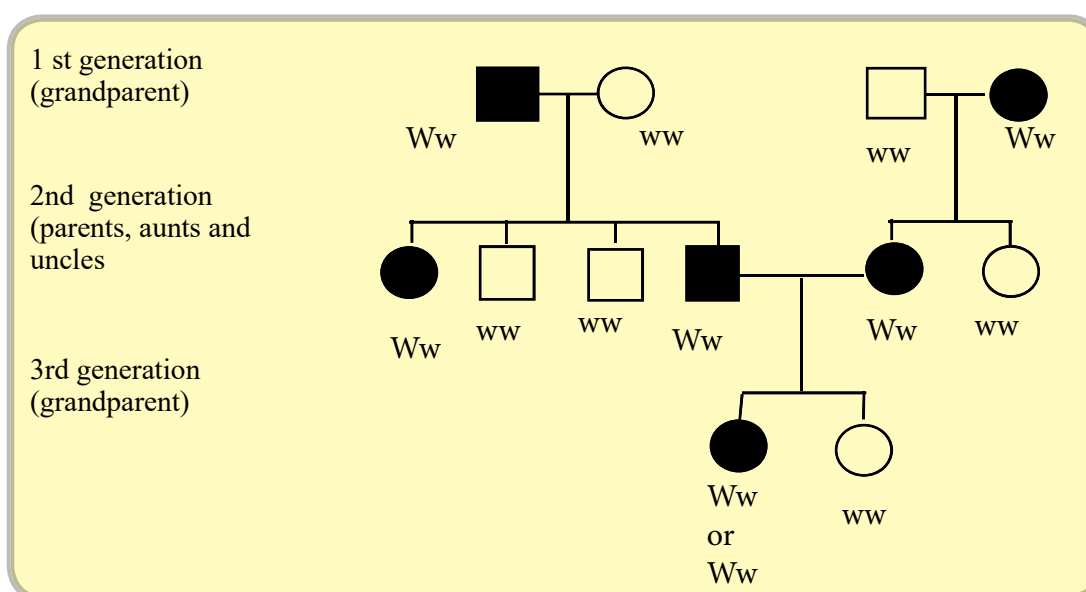


Figure 6.7: Inheritance of Widow's peak

Attached ear lobe

Attached ear lobe, as explained earlier, is a recessively inherited character. In the pedigree chart given below, the inheritance of the trait is analysed in the same family that was used to study the widow's peak. The dominant allele, which is causing the free ear lobe is denoted with 'F' while recessive allele is denoted with 'f'.

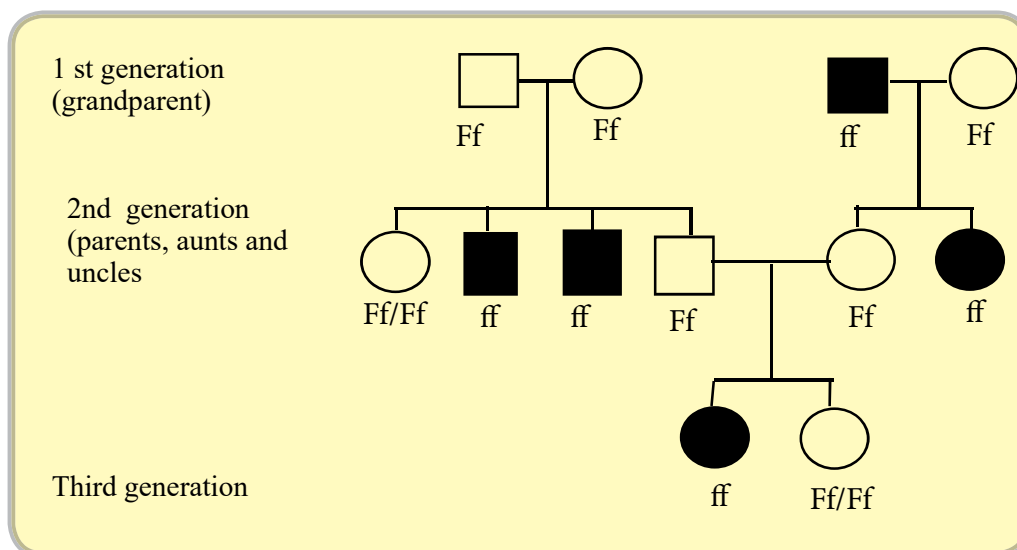


Figure 6.8: The pedigree chart showing the inheritance pattern of attached lobe

In the first generation, both parents lacking the attachment of ear lobes, resulted mixed progenies with attached and free ear lobes. This tells us that those two parents were heterozygous (Ff), and among the progeny, two males with attached ear lobes carry homozygous recessive alleles (ff) and a male and female with free ear lobe may carry heterozygous (Ff) or homozygous dominant alleles (FF or Ff). Another cross happened at the first generation between a male with attached ear lobe and a female with free ear lobe. This resulted one daughter with attached ear lobe. Therefore, she must possess ff and the other may be FF or Ff . A male from one family and a female from another family, at the second generation expressing free ear lobe phenotype, had resulted progeny having two females, one with attached ear lobes and the other with free ear lobes for the third generation. Therefore, the second generation male and female crossed with free ear lobes must be Ff . Third generation female with attached ear lobes must carry ff genotype and the other may carry either FF or Ff .

The probability that another child from the same family will have attached earlobes could be calculated using a monohybrid cross ($Ff \times Ff$). Since homozygous recessive (ff) genotype is causing the condition, the probability is $1/4$ for each child.

The chance that a child born to the family will have both the widow's peak and attached earlobes can be calculated using probability rules. Assuming that the alleles corresponding to the two

characters are on different chromosomes, the two pairs of alleles assort independently in this dihybrid cross (WwFf X WwFf).

Thus according to multiplication rule,

Chance of having both widow's peak (WWff/Wwff)

and attached ear lobe = Chance of widow's peak x chance of attached earlobe

$$= \frac{3}{4} \times \frac{1}{4}$$

$$= \frac{3}{16}$$

Non-Mendelian inheritance

Non-Mendelian inheritance refers to inheritance patterns in which traits do not segregate in accordance with **Mendel's** laws of inheritance. Phenotypes that do not appear in ratios predicted by Mendelian genetics are the indicators of Non-Mendelian inheritance.

Examples for non-mendelian patterns

- when alleles are not completely dominant or recessive (incomplete dominance and co-dominance),
- when a particular gene has more than two alleles (polyallelism)
- when a single gene produces multiple phenotypes (pleiotropy)
- sometimes two or more genes are involved in determining a particular phenotype (epistasis and polygenic inheritance)
- gene linkage
- genes which are located in sex chromosomes exhibit a different pattern of inheritance in males and females due to the unequal distribution of genes in their sex chromosomes.

Incomplete dominance

The phenomenon of dominant allele completely masking the recessive phenotype, resulting similar phenotypes for both homozygous dominant zygote as well as heterozygous zygote is called **complete dominance**.

On the other hand, at the heterozygous state, the phenomenon of expressing blend phenotypes from both alleles is called **incomplete dominance**. Degree of expression of each allele in the blend phenotypes may vary based on the nature of the alleles.

In *Mirabilis jalapa* (Four o'clock plant), there are several types of flower colours. When red flowered plants are crossed with white flowered plants, all the F₁ hybrids (heterozygotes) have pink flowers (Figure 6.8). This third, intermediate phenotype results from flowers of the heterozygotes having less red pigment than the red homozygotes.

When these F_1 pink flowers are self-pollinated or crossed among themselves to raise F_2 generation, they produce red (C^{RR}), pink (C^{RW}) and white (C^{WW}) flowers giving 1:2:1 ratio. This phenotypic ratio is identical with genotypic ratio because heterozygotes are phenotypically intermediate between two homozygous types.

Note: Since, neither allele is dominant, instead of upper- and lowercase letters, a superscript is used to indicate the trait; i.e. C^R for red colour and C^W for white colour.

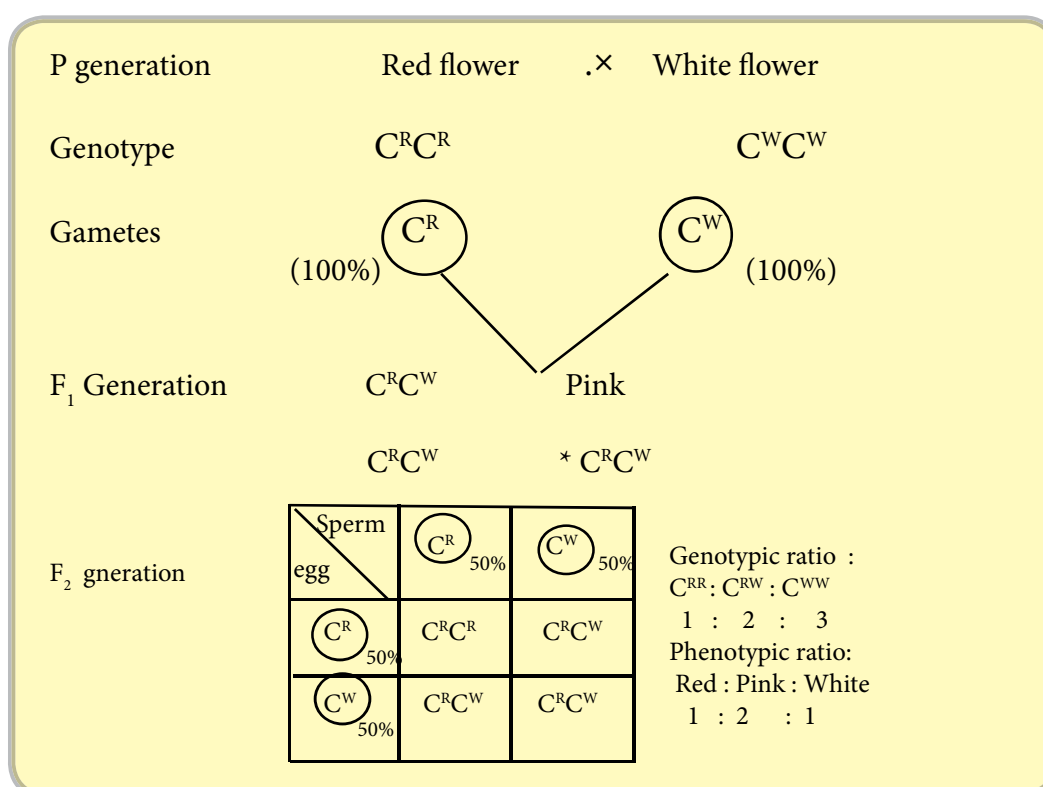


Figure 6.9: Incomplete dominance in *Mirabilis jalapa* flower colour gene

Codominance

In certain traits, at heterozygote state, expression of both alleles contributes equally to the phenotype. Such phenomenon is called **co-dominance**.

For example, a person with AB blood group type has both A and B carbohydrates on the surface of red blood cells at the same time. The two carbohydrates are added to the surface of RBC by enzymes encoded by the I^A and I^B alleles of a single gene. A heterozygous individual would express both carbohydrates ($I^A I^B$) in an equal manner.

Likewise,

Homozygotes for I^A allele ($I^A I^A$) will carry only A carbohydrate on RBC.

Homozygotes for I^B allele ($I^B I^B$) will carry only B carbohydrates on RBC.

As shown below, the F_1 progeny resulting from the mating of two homozygous individuals for each allele would consist of individuals only of AB blood group. F_2 generation produced by the possible mating with in F_1 progeny or heterozygous individuals for the AB alleles, would produce the three phenotypes, blood type A: AB: B at a ratio of 1:2:1.

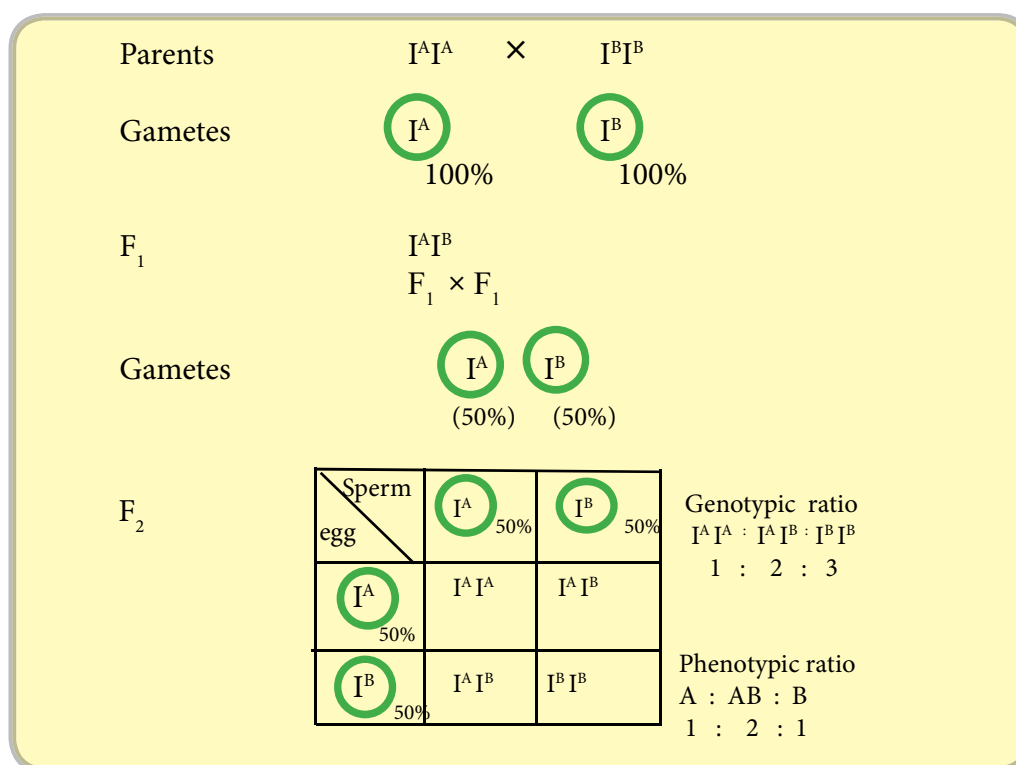


Figure 6.10: Codominance in ABO blood grouping

As shown, the F_2 phenotype ratios in both incomplete dominance and co-dominance are 1:2:1. Hence the two phenomena cannot be differentiated using F_2 phenotypic ratios.

The key to differentiate between incomplete dominance and co-dominance is that in incomplete dominance F_1 generation shows a different phenotype than both the parents whereas in co-dominance F_1 generation shows both the parental traits together.

Polyallelism (Multiple alleles)

Polyallelism refers to the presence of multiple alleles for a single genetic locus, a phenomenon where certain traits are determined by the combination of more than two types of alleles.

E.g. there are three alleles called I^A , I^B , and i for a single genetic locus which at different combinations determine ABO blood groups in humans.

In any diploid individual there are only two of the several alleles are present.

As mentioned earlier, the alleles I^A and I^B code for enzymes that add A and B carbohydrates to the surface of red blood cells. The two alleles are in a co-dominant relationship. However, the allele 'i' results in lack of these carbohydrates on the red cell surface and it is recessive to both I^A and I^B alleles. Therefore, both $I^A i$ and $I^B i$ genotypes will result in dominant phenotypes; i.e. having either the A or B carbohydrates. The ii genotype will result in the recessive trait of not having either of the carbohydrates.

Thus, based on the presence of the two carbohydrates, a person's blood group may be one of four types as follows: type A (carbohydrate A present), type B (carbohydrate B present), type AB (both carbohydrates A & B present), or type O (neither of the two carbohydrates present).

F_1 and F_2 generations resulting from a mating between individuals with different blood groups are shown below

Parents		$I^A I^A$	×	ii			$I^B I^B$	×	ii
Gametes		I^A		i			I^B		i
		100%		100%			100%		100%
F_1		$I^A i$				$I^B i$			

F_2		$I^A i \times I^B i$			
Sperm	egg	I^A	i	F_2 Genotype ratio	
		50%	50%	$I^A I^B : I^A i : I^B i : ii$	
	I^B	$I^A I^B$	$I^B i$	1 : 1 : 1 : 1	
	i	$I^A i$	ii	F_2 phenotype ratio	
				AB : A : B : O blood type	
				1 : 1 : 1 : 1	

Figure 6.11: Multiple alleles for the ABO blood groups

Epistasis

Epistasis is the phenomenon resulting from interactions between genes of different loci. The alteration in the phenotypic expression of a gene at one locus is due to the interference of another gene at a different locus.

Epistasis could be categorized into two types; dominant epistasis and recessive epistasis, based on the nature of the gene interaction. This causes deviation from the phenotype ratios of Mendelian principles.

Dominant epistasis

When a dominant allele at a specific locus alters the expression of a separate gene at a different locus, it is referred to as **dominant epistasis**. Dominant epistasis is seen in the plumage colour of house fowls. Consider the following experiment.

A cross between a known homozygous, double dominant, white housefowl and a homozygous double recessive white house fowl results a 100% white F_1 progeny. F_2 generation produced from a cross between these F_1 individuals consists of white and coloured fowls in the ratio of 13:3.

The colour of these two fowl varieties are determined by two separate genes;

- The gene 'C'/'c' is responsible for producing colour in the feathers. The dominant 'C' allele produces colour while the recessive 'c' allele result in the absence of pigmentation.
- The gene 'I' is epistatic to gene 'C' and suppresses the expression of the colour. The dominant 'I' allele prevents production of pigments while recessive 'i' allele is unable to prevent colouration.

As a result, the double dominant homozygous (CCII) house fowl is white (dominant I allele prevents colouration). The double homozygous recessive house fowl (ccii) is white (recessive c allele couldn't produce colour).

The F_1 generation all consist of heterozygous (CcIi) fowls. Due to the inhibitory effect of the dominant 'I' allele, all F_1 individuals are white. When these fowls are allowed to interbreed, the F_2 generation genotypes carrying inhibitory 'I' allele will give rise to white colour plumage, despite the presence of the dominant 'C' allele. The presence of dominant 'C' allele in the absence of inhibitory 'I' allele gives rise to coloured fowls.

P:	CCII	×	ccii
	(White house fowl with double dominant alleles)	↓	(White house fowl with double recessive alleles)
F ₁	CcIi		
	(white)		
	F ₁ × F ₁		
	CI	Ci	cI
	CCII	CCIi	CcII
CI	(white)	(white)	(white)
	CCIi	CCii	CcIi
Ci	(white)	(coloured)	(white)
	CCII	CcIi	ccII
cI	(white)	(white)	(white)
	CcIi	Ccii	ccIi
ci	(white)	(coloured)	(white)

Phenotype ratio white : colour

13 : 3

Figure 6.12: An example of dominant epistasis-Plumage colour of house fowl

As per the Mendelian principles, genotype ratios in both F₁ and F₂ generation are expected to be similar to the genotype ratios seen in a normal dihybrid cross. However, due to the effect of epistasis, phenotype ratio would deviate from Mendel's principles for normal dihybrid cross. F₂ phenotypic ratio may get altered from 9:3:3:1 to 13:3.

Recessive epistasis

When a homozygous recessive genotype of a particular chromosomal locus alters/ masks the expression of a separate gene at a different locus, it is referred to as **recessive epistasis**.

A good example for recessive epistasis is found for flower colour in sweet pea (*Lathyrus*) plant. There are purple flowered and white flowered varieties.

A cross between homozygous dominant purple flower (AABB) and homozygous recessive white flower (aabb) strains produced 100% purple colour flowers in F₁ generation. Inter breeding of F₁ plants produced F₂ generation with purple and white flower plants in a ratio of 9: 7.

The purple colour in sweet pea flower is governed by two dominant genes, A and B. Both A and B alleles are coding for compounds that are necessary for the expression of purple colour. Hence, purple colour will be there only when both the dominant alleles (A and B) are present.

Double recessive genotype at any locus (AAbb, aaBB, Aabb or aabb) results white flowers by masking the expression of purple colour. Thus, double recessive genotype at any of the above locus is epistatic to either homozygous dominant (AA and BB) or heterozygous condition (Aa and Bb) of the other. (AAbb, Aabb, aaBB, aaBb, aabb- White and AaBb, AaBB, AABb, AABB- Purple)(Figuer 6.13)

All of the F_1 generation plants were found to express purple flowers due to the presence of heterozygous condition at both loci (AaBb).

In F_2 generation, plants having genotypes with both **A** and **B** alleles (9/16) express purple flowers, and plants having genotypes with '**aa**' and a '**B**' allele (3/16) or '**A**' allele and '**bb**' alleles (3/16) and '**aabb**' genotype (1/16) produce white flowers, thus only two phenotypic classes are expressed; purple and white. Thus, the normal dihybrid phenotype ratio as per Mendelian principles **9:3:3:1** is changed to 9: 7 ratios in F_2 generation.

Parents	Purple Flower AABB	×	White Flower aabb
		↓	
F_1	AaBb		(Purple Flower)
$F_1 \times F_1$	F_2		
	AB Ab	aB ab	
AB	AABB AABb	AaBB AaBb	
	[P] [P]	[P] [P]	
Ab	AABb AAbb	AaBb Aabb	
	[P] [W]	[P] [W]	
aB	AaBB AaBb	aaBB aaBb	
	[P] [P]	[W] [W]	
ab	AaBb Aabb	aaBb aabb	
	[P] [W]	[W] [W]	
P = Purple Flower, W = White Flower			

Figure 6.13: an example of recessive epistasis - flower colour of sweat pea plants

Polygenic inheritance

Inheritance of a phenotype such as quantitative characters; height, skin colour, intelligence quotient etc. which results from a cumulative expression of two or more genes is called polygenic inheritance.

e.g: Skin colour in humans is determined by many genes. For simplicity only three genes are considered. Each gene (A, B, or C) has a dark-skin allele contributing one “unit” of darkness to the phenotype and being incompletely dominant to the other allele (a, b, or c). Therefore

AABBCC person: very dark skin

aabbcc person: very light skin

AaBbCc person: intermediate between skin colour

Because the dominant alleles from various loci have a cumulative effect, on the skin colour. (Campbell et al, 2015)

Based on the number of genes involved in determining a polygenic character, the phenotypic and genotypic combinations in the progenies may vary. Data for a polygenic character representing a population may result in a **normal distribution curve**. The majority of offspring would be expected to have intermediate phenotypes (skin colour in the middle range).

Genetic linkage

Some genes coding for particular characters are located on the same chromosome and also at a closer distance. Thus, they escape from crossing over and independent assortments occur during the meiotic cell division at gametogenesis and inherit together. This results in deviation from Mendel's law of independent assortment. The above phenomenon is called genetic linkage.

E.g. Inheritance of body colour and wing size in the fruit fly *Drosophila*

In *Drosophila*, wild-type flies are found to have gray bodies and normal-sized wings. Due to mutation for the above traits body colour becomes black and wings become vestigial. Both characters are determined by genes of autosomal chromosomes.

In this example, the mutant alleles are recessive to the wild-type alleles. The alleles for body colour are indicated as G (gray) and g (black), and those for wing size are indicated as N (normal) and n (vestigial).

To examine the above, wild type flies were crossed with flies which are mutant to both body color as well as wing size and followed by a dihybrid test cross.

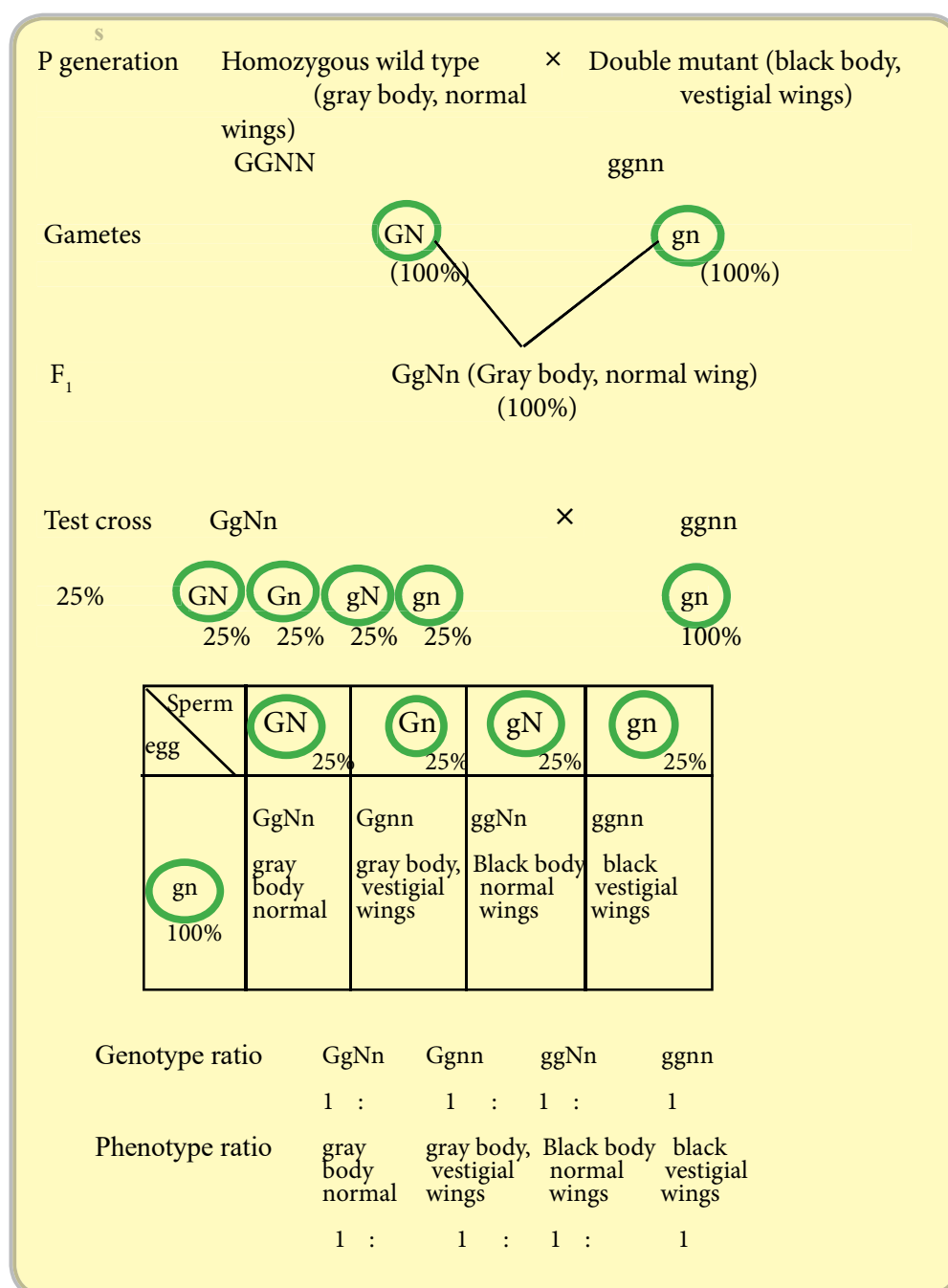


Figure 6.14: Prediction of inheritance of body colour and wing size in the fruit fly *Drosophila* as per Mendelian principles

Most offspring had a parental (P generation) genotype, indicating that the genes for body colour and wing size are genetically linked on the same chromosome.

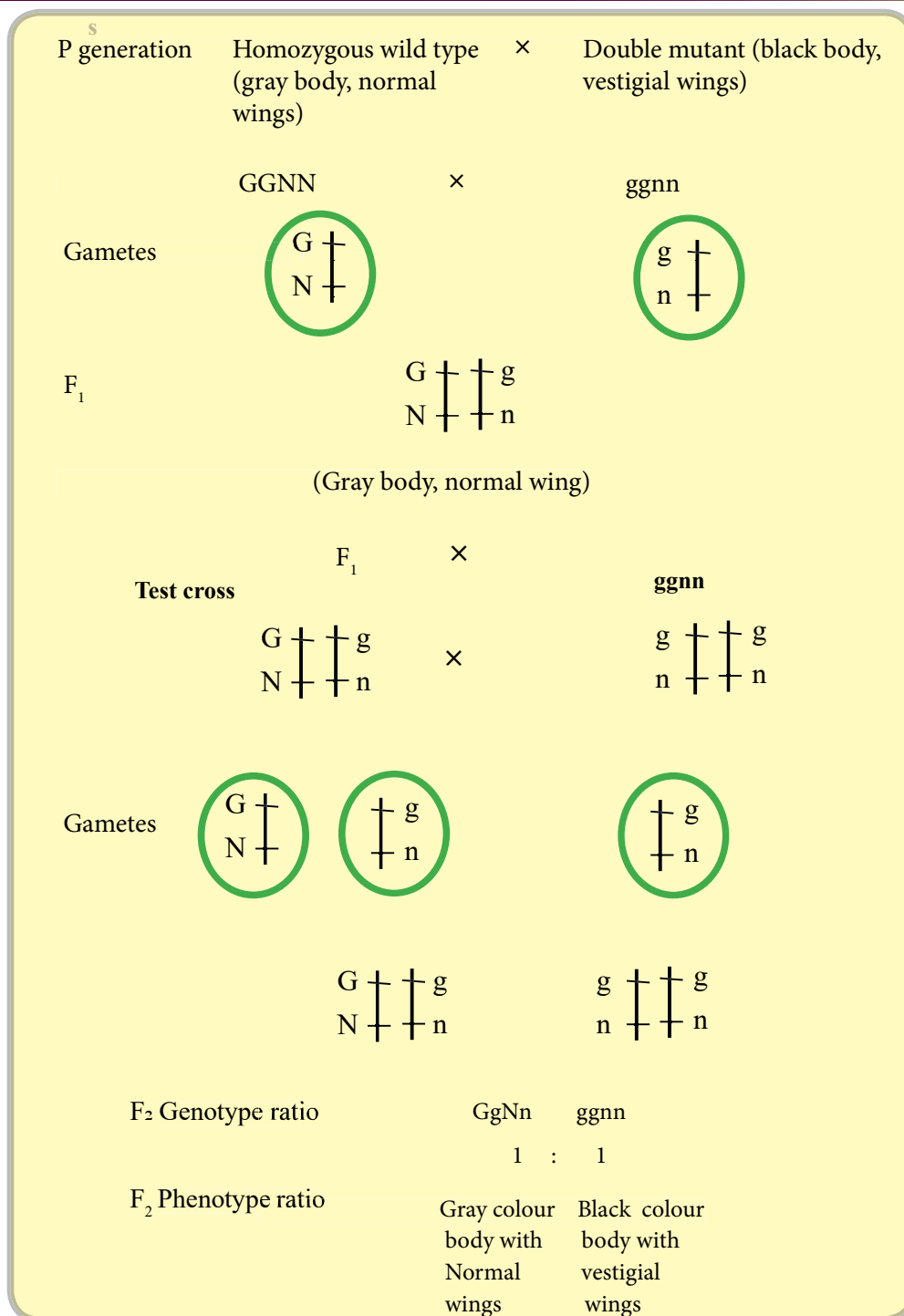


Figure 6.15: Inheritance of body colour gene and wing size gene in the fruit fly *Drosophila*

Though, the genes for body colour and wing size are linked, in some occasions, they get assorted independently due to crossing over. Therefore, the above test cross may results recombinant offsprings in lower frequency. for example, in Morgan's experiment recombinant phenotypes Gray colour body with vestigial wings (Ggnn) and Black colour body with normal wings (ggNn) were observed in lesser numbers.

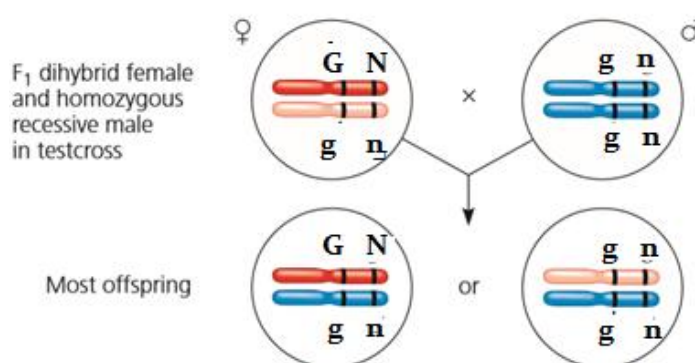


Figure 6.16: Linkage of genes responsible for body colour and wing size in the fruit fly *Drosophila*

The production of a relatively small number of offspring with non-parental phenotypes indicated occasional breaks in the genetic linkage. This is due to the crossing over occurs between the homologous chromosomes.

Human sex determination

Sex is determined by the expression of sex chromosomes. In humans, each individual carries 22 pairs of autosomal chromosomes and one pair of sex chromosomes. Type of sex chromosomes expressing male traits are named Y chromosome and the other as X chromosome. Comparatively X chromosome is bigger than Y chromosome. They both code for different traits except their homologous regions. When X and Y chromosomes pair up, they remain homologous only in specific regions. On the other hand when chromosome X, X pair up, they both remain homologous to each other.

On the occurrence of gametogenesis in females, meiosis yields haploid eggs carrying 100% X chromosomes, whereas in males, half the number of the haploid sperms produced carry X and the remaining half Y chromosomes. During the fertilization of male and female gametes, the occasion where both egg and sperm carry X chromosomes results in a female zygote, and on the other hand, an occasions where an egg fuses with a sperm carrying Y chromosome results in a male zygote.

Thus, any mating occasions between male and female organisms of same species leading to fertilization can have fifty percent chance for yielding either male or female zygotes.

Anatomical sex signs develop in humans based on the expression of XX and XY chromosomal combination.